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CLINICAL DESK REFERENCE FOR STUDENTS AND PRACTICING VETERINARIANS

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Foreword

It is a matter of pride and privilege that the Veterinary Clinical Complex and Department of Veterinary Clinical Medicine, Veterinary College and Research Institute, Namakkal have prepared the book "Clinical Desk Reference for the students and practising veterinarians". This book is a comprehensive resource designed for use in educational settings, as well as veterinary practitioners

This book provides veterinarians with detailed information in field practice. As the veterinary profession becomes increasingly diversified, there is a greater need for a textbook that covers all aspects of veterinary medicine as it relates to day-to-day field practice.

The editors have tried to compile the relevant and useful information on practical aspects for daily practice. All of the information is organised in a way that is designed to enable the reader to look up topics quickly and at their convenience. Students as well as practising veterinarians will find this book extremely useful.

I am confident that this book will prove to be a useful aid for the students and veterinary practitioners and will be a ready reckoner for them. This will surely help in arriving at diagnosis and treatment effectively and efficiently.

I would like to congratulate the Alembic pharmaceuticals for the publication of this book. I wish them good luck for all their endeavours.

(T. SATHIAMOORTHY)
01/06/22

PREFACE

"The greatness of a nation and its moral progress is judged by the way its animals are treated"
- Mahatma Gandhi

The *"Clinical Desk reference for students and practicing veterinarians"* is a one stop destination for all your day-to-day clinical practice. From normal vital parameters, haemato-biochemistry, drug doses and dosage intervals to drug contraindications, proformas, field test and more, this book will answer all your questions related to daily practice.

This book is a handy guide for students and practicing veterinarians as they will find it useful to serve as a quick reference.

The authors invite any kind of suggestions, corrections, additions and alterations that may remain within the text to improve it in future.

The authors express sincere gratitude to all their teachers for their inspiration and valuable guidance.

The completion of the work would have not been possible in time without the efforts of the entire staff members of alembic Pharmaceuticals.

Authors

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CHAPTER 1

NORMAL VITAL PARAMETERS IN VARIOUS SPECIES

SPECIES	TEMPERATURE (°C)	PULSE RATE (Rates/min)	RESPIRATORY RATE (Breaths/min)
Cattle (above 1 year)	37.8-39.2	60-72	10-30
Calf (upto 1 year)	38.5-39.5	80-120	24-36
Sheep	38.5-40.0	70-90	20-30
Goat	38.6-40.2	70-90	20-30
Horse	37.5-38.5	28-46	8-16
Pig	37.8-38.5	60-90	10-20
Dog (Small breed)	38.5-39.2	60-140	24-36
Dog (Large breed)	37.5-38.5	Up to 180	18-30
Puppies	37.8-39.3	Up to 220	20-30
Cat	37.8-39.2	120-240	20-30

CHAPTER 2

HAEMATOLOGICAL VALUES IN VARIOUS SPECIES AND ITS INTERPRETATIONS

PARAMETERS	UNITS	CANINE	FELINE	BOVINE	OVINE	CAPRINE	EQUINE
Erythrocytes	($10^6/\mu l$)	5-9	5-10	5-10	9-15	8-18	6.8-12.9
Haemoglobin	(g/dl)	12-19	9.8-15.4	8-15	9-15	8-12	11-19
Hamatocrit	(%)	37-57	29-45	24-46	27-45	22-38	32-53
MCV (Mean corpuscular volume)	(fl)	66-77	41-54	40-60	28-40	16-25	37-59
MCH (Mean corpuscular haemoglobin)	(pg)	21-26	13-17	11-17	8-12	5.2-8	12.3-19.7
MCHC (Mean corpuscular haemoglobin concentration)	(g/dl)	32-36	31-36	30-36	31-34	30-36	31-38.6
Platelets	($10^5/\mu l$)	1.6-5.1	2.3-6.8	1-8	2.5-7.5	3-6	1-3.5
Total leukocytes	($10^3/\mu l$)	5-15	5.5-19.5	4-12	4-12	4-13	5.4-14.3
Neutrophils	%	60-75	35-75	15-45	10-50	30-48	35-75
Lymphocytes	%	18-21	27-36	48-75	40-75	50-70	20-70
Monocytes	%	2-10	0-5	2-7	1-5	1-4	1-8
Eosinophils	%	0-9	0-4	2-15	1-8	3-8	1-10
Basophils	%	0-1	0-1	0-2	0-3	0-2	0-3

<table>
<tr><th>S. No.</th><th>PARAMETERS</th><th colspan="2">INCREASED LEVEL</th><th>DECREASED LEVEL</th></tr>
<tr><td>1</td><td>Erythrocytes</td><td colspan="2">Polycythemia (high altitude, sports animals, myeloid leukemia), Haemoconcentration</td><td>Anaemia (blood loss, impaired RBC production due to deficiency of iron, copper, vitamin B12 and folic acid)</td></tr>
<tr><td>2</td><td>Haemoglobin</td><td colspan="2">Polycythemia, dehydration, haemoconcentration, high altitude</td><td>Anaemia</td></tr>
<tr><td>3</td><td>Hematocrit</td><td colspan="2">Haemoconcentration, polycythemia</td><td>Bone marrow depression, Anaemia</td></tr>
<tr><td rowspan="2">4</td><td rowspan="2">MCV (Mean corpuscular volume)</td><td>Classification</td><td>MCHC Normal</td><td>MCHC Decreased</td></tr>
<tr><td>MCV Normal</td><td>Normocytic Normochromic</td><td>Normocytic Hypochromic</td></tr>
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<tr><td>6</td><td>MCHC (Mean corpuscular haemoglobin concentration)</td><td>MCV Decreased</td><td>Microcytic Normochromic</td><td>Microcytic Hypochromic</td></tr>
<tr><td rowspan="4"></td><td>Microcytic hypochromic anaemia</td><td colspan="3">Dietary deficiency of iron, protein, copper, pyridoxine and chronic blood loss</td></tr>
<tr><td>Normocytic normochromic anaemia</td><td colspan="3">Depressed erythropeisis due to chronic infectious diseases, nephritis, irradiation, malignant neoplasms, chemical poisoning and vitamin E deficiency</td></tr>
<tr><td>Macrocytic hypochromic anaemia</td><td colspan="3">Acute haemorrhage or acute haemolysis and recovery stage of diseases</td></tr>
<tr><td>Macrocytic normochromic anaemia</td><td colspan="3">Deficiency of vitamin B12, cobalt, folic acid, intrinsic and erythrocyte maturation factor</td></tr>
</table>

S. No.	PARAMETERS	INCREASED LEVEL	DECREASED LEVEL
7	**Platelets**	Vigorous exercise (due to mobilizatio of platelets from spleen and lungs), splenectomy, inflammation, myeloproliferative disease.	Ionizing radiation, viral infections (equine infectious anaemia), rickettsial infections (ehrlichiosis), immune mediated reactions (idiopathic thrombocytopenic purpura, autoimmune haemolytic anaemia, systemic lupus erythematosis).
8	**Total leukocytes**	Leukaemia, inflammatory disease, exercise, fear, near term of pregnancy, day of estrus.	Viral diseases, shock
9	**Lymphocytes**	Chronic infection, viral infections, infectious hepatitis, bacillary infections, tuberculosis, goiter, lymphoma, lymphocytic leukaemia.	TB, acute stress, few viral infections, immunosuppressive drugs, radiation.
10	**Monocytes**	Chronic infection, endocarditis, typhoid, brucellosis, TB, subacute bacterial infections, swine Erysipelas, listeriosis, E.canis, monocytic and myelomonocytic leukemia, relative increase in leukopenia, neutropenia, prolonged blood parasitic infections (trypanosomiasis)	
11	**Neutrophils**	Acute infections, bacterial infections, pneumonia, metritis, pyometra, meningitis, rheumatoid arthritis, localized pyaemic infections, like abscesses, furuncle, otitis, intoxication like uraemia, gout, diabetic acidosis, drug poisoning like lead, mercury, insect venoms, tissue necrosis, acute haemorrhage, sudden hemolysis, granulocytic leukaemia, myeloid leukemia, acute bacterial infections,	Viral infection, chronic infections like TB, brucellosis, protozoal and fungal infections. Endotoxaemia, immune mediated diseases, cancer chemotherapeutic drugs, radiation, idiosyncratic drug

S. No.	PARAMETERS	INCREASED LEVEL	DECREASED LEVEL
		septicaemia, toxaemia, hypersplenism, anti-inflammatory drugs (cortisol), antibacterial drugs (chloramphenicol, sulphonamides)	reactions, infectious agents, toxicoses like bracken fern poisoning, estrogen toxicity, myelophthisis, genetic disorders,
12	**Eosinophils**	GI parasitic infections, allergic conditions like bronchial asthma, allergic rhinitis, skin diseases, parasitic diseases, eosinophilic myositis, radiation therapy, tumours of ovary, serous surfaces and bones, eosinophilic leukaemia, poisons like copper sulphate, phosphorus, camphor, drug reactions following penicillin and sulphonamides administration.	Stress condition, administration of ACTH or cortisol, epinephrine treatment, corticosteroids.
13	**Basophils**	Allergic conditions, haematological malignancies, chronic sinus inflammation, chronic haemolytic anaemia, foreign protein infection, Hodgkin's disease, basophilic leukaemia, granulocytic leukaemia.	

CHAPTER 3

SERUM BIOCHEMICAL VALUES IN VARIOUS SPECIES AND ITS INTERPRETATIONS

PARAMETERS	UNITS	CANINE	FELINE	BOVINE	OVINE	CAPRINE	EQUINE
Glucose	(mg/dl)	65-118	73-134	45-75	50-80	50-75	75-115
Cholesterol	(mg/dl)	135-270	95-130	65-220	52-76	80-130	46-180
Total protein	(g/dl)	5.4-7.1	5.4-7.9	5.7-8.1	6.0-7.9	6.4-7.0	6-7.7
Albumin	(g/dl)	2.3-3.3	2.1-3.9	2.1-3.6	2.4-3.0	2.7-3.9	2.9-3.8
BUN (Blood urea nitrogen)	(mg/dl)	8-28	19-34	6-27	8-20	10-20	10-24
Creatinine	(mg/dl)	0.5-1.4	0.8-2.2	1-2	1.2-1.9	1-1.8	0.9-1.9
Total Bilirubin	(mg/dl)	0.10-0.5	0.15-0.5	0.01-0.5	0.1-0.5	0-0.1	1-2.0
Conjugated bilirubin	(mg/dl)	0.06-0.12	-	0.04-0.44	0-0.27	-	0-0.4
ALT (Alanine amino transferase)	(IU/L)	21-102	6-83	11-40	5-20	24-83	3-23
AST (Aspartate amino transferase)	(IU/L)	13-66	26-43	78-132	60-280	167-513	226-366
ALP (Alkaline phosphatase)	(IU/L)	20-156	2-93	0-200	70-390	93-387	140-400
GGT (Gammaglutamyl transferase)	(IU/L)	1-6.4	1.3-5.1	6.1-17.4	20-52	20-56	4.3-13.4
CK (Creatine kinase)	(IU/L)	52-368	69-214	35-280	-	-	145-380
Amylase	(IU/L)	185-700	550-1458	-	-	-	75-150
Lipase	(IU/L)	60-330	0-76	<50	-	-	<20
Calcium	(mg/dl)	9-11.3	6.2-10.2	9.7-12.4	11.5-13	8.9-11.7	11.2-13.6
Phosphorus	(mg/dl)	2.6-5.3	2.6-6.1	5.6-6.5	5-7.3	4.2-9.1	3.1-5.6
Sodium	(mmol/L)	141-152	146-156	132-152	145-152	142-155	132-146
Potassium	(mmol/L)	3.9-5.1	3.7-6.1	3.9-5.8	3.9-5.4	3.5-6.7	3-5
Magnesium	(mmol/L)	1.8-2.4	1.7-2.6	0.74-1.10	0.90-1.26	0.31-1.48	0.9-1.2
Chloride	(mmol/L)	105-115	115-130	95-110	95-103	99-110.3	98-110
TT4	(μg/dl)	1-4	0.8-4.7	4.2-8.6	-	-	1-3.8
Cortisol	(μg/dl)	0.96-6.81	0.33-2.57	0.47-0.75	1.40-3.10	-	2-6

S. No.	PARAMETERS	INCREASED LEVEL	DECREASED LEVEL
1	**Glucose**	Ketosis, pregnancy toxemia, diabetes mellitus, Hyperactivity of thyroid, pituitary & adrenal gland, Pancreatitis, asphyxia, anesthesia, pneumonia, dehydration, certain hepatic disorders, drug & toxin induced hyperglycemia (glucocorticoids, progestogens, α2 receptor agonist, ß blockers, glucose containing crystalloids, ethylene glycol toxicity)	Hypothyroidism, Hypopituitarism, Hyperinsulinism, hypoadrenocorticism, Liver diseases, over dosage of insulin, physiological condition –pregnancy, starvation, severe prolonged exercise & lactation.
2	**Cholesterol**	Hypothyroidism, hyperadrenocorticism, pancreatitis, obesity pregnancy, Nephrotic syndrome, biliary cirrhosis, athesclerosis, Diabetes	Hyperthyroidism, Malnutrition, Gaucher's disease, liver diseases, exocrine pancreatic insufficiency, histiocytic sarcoma, multiple myeloma, lymphoma, protein losing enteropathy
3	**Total protein**	Severe dehydration, High total protein with normal albumin- multiple myeloma, plasma cell tumors, lymphocytic leukemia, chronic autoimmune disorders (systemic lupus rheumatoid arthritis)	Nutritional hypoproteinemia, malabsorption, liver diseases, sepsis, Renal diseases & nephrotic syndrome
4	**Albumin**	Severe dehydration, hepatocellular carcinoma	Hepatic failure, Malnutrition, severe burns, protein losing glomerulopathy, severe exudative dermatopathies, Auto immune diseases, gastro intestinal protein losing enteropathies, hypoadrenocorticism, hyperglobulinemia, crohn's diseases, Hodgkin's lymphoma, uncontrolled diabetes, Hyperthyroidism, Heart failure,

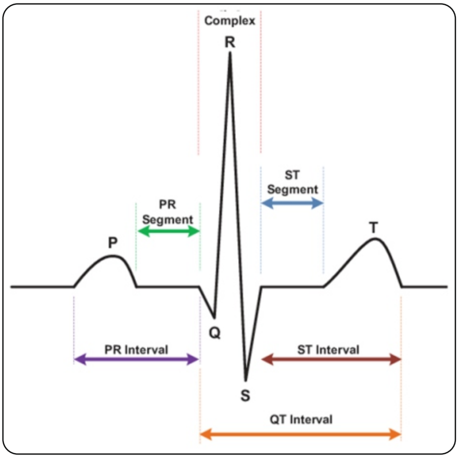
S. No.	PARAMETERS	INCREASED LEVEL	DECREASED LEVEL
5	**BUN (Blood urea nitrogen)**	Increased protein catabolism, increased protein digestion due to haemorrhage and high protein diet Decreased GFR • Pre renal – poor renal blood flow link in shock or renal stenosis. • Renal – damage to renal parenchyma. • Post renal – obstruction in kidney / urinary tract increases the tubular reabsorption of urea.	Protein malnutrition, liver diseases, increased excretion during polyuria due to hyperadrenocorticism, diabetes mellitus
6	**Creatinine**	Renal diseases, extensive muscle destruction (not influenced by increased protein diet)	Decreased muscle mass, increased GFR
7	**Bilirubin**	Anorexia, fasting, hemolytic anemia, Blood levels of conjugated (direct) bilirubin are elevated with hepato cellular damage (bile duct injury/ obstruction). Blood levels of unconjugated bilirubin are elevated (excessive erythrocyte destruction / defects in the transport mechanism that allows bilirubin to enter hepatocytes for conjugation)	
8	**ALT (Alanine amino transferase) or SGPT (Serum glutamic pyruvic transaminase)**	Hepatic necrosis, hepatocellular carcinoma, cholestasis, muscle injury in large animals, hyperthyroidism in cats	
9	**AST (Aspartate amino transferase) or SGOT (Serum glutamic oxalo-acetic transaminase)**	Nonspecific liver damage, muscle inflammation/ necrosis, spontaneous/ artifactual hemolysis), hyperthyroidism in cats	

S. No.	PARAMETERS	INCREASED LEVEL	DECREASED LEVEL
10	**ALP (Alkaline phosphatase)**	Increased in cholestasis (only used in dogs & cats), exogenous corticosteroids in dogs, hepatocellular/ biliary neoplasia, hyperadrenocorticism, increased osteoblastic activity	
11	**GGT (Gamma glutamyl transpeptidase)**	Biliary hyperplasia, cholestasis, corticosteroids and phenobarbitone therapy in dogs, hyperadrenocorticism	
12	**Creatinekinase / Creatinephos-photase**	Elevated with muscle damage, intramuscular injection, persistent recumbency, surgery, vigorous exercise, electric shock, laceration, bruising, hypothermia, immune mediated polymyositis, neurologic injury	
13	**Amylase**	Acute pancreatitis, flare ups of chronic pancreatitis, obstruction of pancreatic ducts, chronic renal insufficiency, decreased GFR, moderate elevations in intestinal disease/ obstruction	
14	**Lipase**	Acute pancreatitis, hyperadrenocorticism	
15	**Calcium**	Young animlas, thiazide diuretics, Hyperparathyroidism, hypervitaminosis D, multiple myeloma, neoplastic diseases of bone, hypoadrenocorticism, hyperalbuminemia, endometritis, retained fetus	Milk fever, eclampsia, Hypoparathyroidism, hypoproteinemia, hypoparathyroidism, rickets, osteomalacia, steatorrhea, tetany, Renal failure.
16	**Phosphorus**	Hypervitaminosis D, hypoparathyroidism, Renal failure, hyperthyroidism	Diuretics, corticosteroids, alkalemia, hepatic lipidosis, Post parturient haemoglobinuria, osteomalacia, Rickets, pica, hyperparathyroidism, Fanconi syndrome.
17	**Sodium**	Excess water loss, sodium containing intravenous fluids, sodium feed, sodium containing enemas, oral salt wash, Hyperaldosteronism, diabetus insipidus, Hyperventilation , Extensive skin denudation, water deprivation, CNS disease.	Diuretic therapy, congestive heart failure, hepatic disease, hypoadrenocortisim, hypoaldosteronism, decreased intake

S. No.	PARAMETERS	INCREASED LEVEL	DECREASED LEVEL
18	**Potassium**	Chronic kidney disease, uroperitoneum, uroabdomen, hypoadrenocorticism, hypoaldosteronism, hyperkalemic myopathy, diabetes mellitus, hyperchloremic metabolic acidosis	Metabolic alkalosis, hyperinsulinemia, abomasal stasis, Hyperadrenocorticism, Alkalosis, Diuretics, Prominent feature-Apathy, muscle weakness,
19	**Magnesium**	Myopathy, soft tissue necrosis, hyperparathyroidism, occur in herbivores with renal failure, administration of antacids / magnesium cathartics to patients with renal failure.	Grass tetany, wheat pasture poisoning, hypoalbuminemia, hyperaldosteronism, hyperthyroidism, primary hypoparathyroidism
20	**Chloride**	Hyperchloridemia – in dehydration, congestive heart failure, decreased renal blood flow, Iatrogenic, metabolic acidosis	Hypochloridemia – losses from GI tract (vomiting), abomasal displacement, gastric rupture, Advanced renal failure, Adrenal insufficiency, Exudation from burns/ wounds

CHAPTER 4

NORMAL ELECTROCARDIOGRAPHIC VALUES



Horse (msec)	Dog (msec)	Cat (msec)	Cattle (msec)	Horse (msec)
P wave	Amplitude: 0.04 mv Duration: 0.04 sec	0.2 mv 0.04 sec	80±10	100±32
PR interval	0.06 - 0.13 sec	0.04 – 0.09 sec	200±20	136±7
Q wave	0.5 mv	-	-	-
R wave	2.5 mv - small breeds 3 mv - large breeds	0.9 mv	-	-
S wave	0.35 mv	-	-	-
QRS duration	0.05 - 0.06 sec	<0.04 sec	60±10	91±10
QT segment	0.15 - 0.25 sec	0.12 – 0.18	370±30	485±52
ST segment	>0.2 mv - elevation	-	-	-
T wave	1/4th of the R wave	0.3 mv	90±10	-
MEA	+40° to +100°	0° to +160°	-	-

CHAPTER 5

DRUG DOSES AND DOSAGE INTERVAL

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Acepromazine	0.02 -0.1mg/kg (IM,SC,IV) 0.56-2.25 mg/kg, q 6-8hr (PO)	0.02 -0.1mg/kg (IM,SC,IV) 1-2 mg/ kg, q8-12 h (PO)	0.01-0.02 mg/kg (IV) (SD) 0.03-0.1mg/kg (SD) (IM)	0.044-0.088mg/kg (SD) (IM,IV,SC)
Acetaminophen	15mg/kg, q8hr (PO)	Not recommended	-	-
Acetazolamide	5-10mg/kg (glaucoma), q8-12h (PO) 4-8mg/kg (other diuretic used), q8-12hr (PO)		-	2.2 mg/kg, Q 6-12 hr (PO)
Acetylcysteine	Antidote: 140mg/kg (loading dose) then 70 mg/kg, q4hr for five doses (PO, IV) Eye:2% solution, q2hr (Topical)	-	-	8 g ,TD (for retained meconium) (SD) (PR)
Acetylsalicylic acid (Aspirin)	Mild analgesia:10 mg/kg, q2hr Anti-inflammatory:20-25 mg/kg, q12 hr Antiplatelet:5-10mg/kg, q24-48hr	81mg/cat, q48hr (PO)	50-100 mg/kg, q 12 hr (PO)	10-20 mg/kg, q48 hr (PO)
ACTH	Response test: collect pre-ACTH sample and inject 2.2 IU/kg (IM) Collect Post-ACTH sample in 2 hr in dogs and at 1 and 2 hr in cats	-	-	-
Activated charcoal	1-4gm/kg (granules) (PO) 6-12ml/kg (suspension)	-	-	-
Adequan	4.4 mg/kg, twice weekly for upto 4 weeks (IM)	-	-	-

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Albendazole	25-50mg/kg q12 hr for 3 days (PO) For giardia use 25 mg/kg, q12 hr for 2 days (PO)	-	Cattle, goat: 10mg/kg (SD) (PO) Sheep: 7.5mg/kg (SD) (PO)	25-50 mg/kg, SD-12hr (PO)
Albuterol	20-50 micro gram/kg, q 6 -8hr (PO) Upto maximum of 100 microgram/kg, q 6-8h (PO)	-	-	0.001-0.008 mg/kg, q4-8 hr (IH)
Allopurinol	10mg/kg, q 8hr Then reduce to 10 mg/kg, q 24 hr	-	-	-
Aluminium carbonate gel	10-30 mg/kg, q8hr (with meals) (PO)	-	-	-
Aluminium hydroxide gel	10-30 mg /kg, q 8hr (with meals (PO)	-	15-60 mg/kg , q8-24h (PO)	60 mg/kg, q 6-8h (PO)
Amikacin	15-30 mg/kg, q 24 hr (IV,SC,IM)	10-14 mg/kg, q 24 hr (IV,SC,IM)	Not recommended	Foals: 22mg/kg, q 24hr (IM, IV) Adult: 10mg/kg, q 24hr (IM, IV)
Aminopentamide	0.01-0.03 mg/kg, q 8-12 hr (IM,SC,PO)	0.1 mg/cat, q 8-12 hr (IM,SC,PO)	-	-
Aminophylline	10 mg/kg, q 8hr (PO,IM,IV)	6.6 mg /kg, q12 hr (PO)		5-12mg/kg, q8-12hr (PO)
Amiodarone	Start with 15 mg/kg loading dose, then 10 mg/kg/day thereafter			Intravenous infusion of 5mg/kg/hr for 1 hr, then 0.8 mg/kg/hr for 23hrs, then 1.9mg/kg/hr for atrial fibrillation
Amitraz	10.6 ml/7.5 litre water. Apply 3-6 topical treatment for 2weeks. For refractory cases this dose has been exceeded to produce increased efficacy. Doses that		Goats: 11ml of 19.9% sol diluted in 7.5 liters (Topical)	Not recommended

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
	have been used include 0.025%, 0 05%, 0.1% concentration applied twice per week and 0.125% solution applied to one half body everyday for 4 weeks to 5 months			
Amitriptyline	1-2mg/kg Range: 0.25-4mg/kg, q12-24 h (PO)	2-4mg/cat /day For cystitis: 2mg/kg /day (2.5-7.5 mg/cat/day) (PO)		
Amlodipine besylate	2.5mg/dog or 0.1mg/kg, Once daily (PO)	0.625mg/cat/day initially and increase if needed to 1.25 mg/cat/day (Average is 0.18mg/kg) (PO)		
Ammonium chloride	100mg/kg , q 12hr (PO)	800mg/cat (approximately 1/3 to ¼ tsp) mixed with food daily (PO)	50-200mg/kg, q12-24hr (PO)	60-520mg/kg, q24hr (PO)
Ammonium molybdate			50-200 TD, q24hr (PO)	
Ammonium tetrathiomolybdate			1.7-3.4 mg/kg, q48h (IV, SC) 3 treatments	
Amoxicillin trihydrate	6.6-20mg/kg, q8-12 hr (PO)	6.6-20mg/kg, q8-12 hr (PO)	11-22mg/kg, q12-24hr (SC)	6-22 NR,q6-12hr (IM)
Amoxicillin/claulanic acid	12.5-25 mg/kg PO, q12h or q8hr for gram-negative infections (PO)	62.5mg/cat , q 12 h or q 8h for gram negative infections (PO)		15-25mg/kg q6-8h, (IV)
Amphotericin B	0.5 mg/kg to a cumulative dose of 4-8mg/kg, Q 48 h, IV (slow infusion)			0.3-0.6 mg/kg, q24-48h (Dilute & give slow IV)
Amphotericin B (liposomal)	2-3mg/kg for 9-12 treatments for a cumulative dose of 24-27mg/kg, 3 times per week (IV)	1 mg/kg for 12 treatments, 3 times /week (IV)		
Ampicillin	10-20mg/kg (ampicillin sodium), q 6-8h (IV,IM,SC) 20-40mg/kg, q8h (PO)	10-20mg/kg (ampicillin sodium), q 6-8h (IV,IM,SC) 20-40mg/kg, q8h (PO)	22mg/kg, q12h (SC, IV)	10-50mg/kg, q6-8h (IM, IV)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Ampicillin+sulbactam	10-20mg/kg, q 8 h (IV,IM)			
Ampicillin trihydrate	10-50mg/kg, q 24 h (SC,IM)	10-20mg/kg, q 24 h (SC,IM)	4-22mg/kg, q12-24hr (IM,SC)	10-22 mg/kg,NR,q6-8hr (IM,PO)
Amprolium	1.25 gm of 20% amprolium powder to daily feed or 30 ml of 9.6% amprolium solution to 3.8 L drinking water for 7 days (PO)			
Amprolium hydrochloride			Calves: 5-10 mg/kg Lambs: 15mg/kg Kids: 50mg/kg q24hr (PO)	Not recommended
Apomorphine hydrochloride	0.03-0.05 mg/kg (IV IM) or 0.1 mg/kg (SC) or Instill 0.25 mg in conjunctiva of eye (dissolve 6mg tablet in 1-2 ml of saline)			
Apramycin sulfate			Calves: 20-40mg/kg, q24hr (PO)	
Ascorbic acid	100-500mg/ animal/day (diet supplement) or 100mg/animal (urine acidification), q8h		Calves: 3g (TD) (SD) (SC)	30mg/kg, q12-24hr,SD (IV) 1000-2000mg/kg q24h (PO) in red maple poisoning
L-Asparaginase	400 U/kg or 10,000 U/ m^2, Weekly for 3 wks (IV,IP,IM)			
Atenolol	6.25-12.5mg/dog, q12h (PO)	6.25-12.5 mg/cat, q12h. Approximately 3mg/kg (PO)		
Atipamezole	Same as meditomidine		0.02-0.1 mg/kg (SD) (IV)	0.05-0.1mg/kg, (SD) (IV)
Atracurium	0.2mg/kg followed by 0.15mg/kg, Q30min (IV)		Sheep: 0.5 mg/kg then 0.2 mg/kg effect (SD) (IV)	0.15 mg/kg then 0.06-0.2 mg/kg, (SD or to effect) (IV)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Atropine	0.02-0.04 mg/kg, q6-8h (IV, SC, IM) OPC and carbamate poisoning: 0.2-0.5mg/kg		0.06-0.12 mg/kg (preanaesthetic) (IV,IM,SC) 0.5 mg/kg (OPC toxicity) (q4h) (IV,IM,SC)	0.001-0.003 (bronchodilation) (SD) (IV) 0.22mg/kg (OPC toxicity) as needed (IV,IM,SC)
Auranofin (triethylphospihine gold)	0.1-0.2mg/kg, q12h (PO)			
Aurothioglucose	1mg/kg/wk for maintenance <10kg: 1mg 1st week 2mg – 2nd week (IM)	0.5-1mg/cat, q7days (IM)		1mg/kg, q 7days (IM)
Azathioprine	2mg/kg PO q24h initially then 0.5-1mg/kg, q48 hr (PO)	0.3mg/kg, q48h (PO)		2-5mg/kg (loading dose) then every 24h (PO)
Azlocillin				25-75mg/kg, q6-12 hr (IV)
Azithromycin	10mg/kg, q48h (PO) 3.3 mg/kg SD	5-10mg/kg (PO) every other day		10mg/kg, q24h for 5days then q48h (PO)
Bacampicillin sodium				20mg/kg, q12h (PO)
Baquiloprim/ Sulfadimidine			40-80mg/kg, q48h (PO)	
Benztropin				2-4mg/kg, q12hr (IM)
Benazepril	0.25-0.5mg/kg, q24h (PO)	0.5-1mg/kg, q24h (PO) or 2.5 mg/cat/day upto maximum of 5 mg/cat/day		
Betamethasone	0.1-0.2mg/kg, q12-24h (PO)			0.02-0.1mg/kg, q24hr (IM,PO)
Bethanechol	5-15mg/dog, q8h (PO)	1.25-5mg/cat, q8h (PO)	0.07mg/kg,q8hr (SC)	0.05-0.75mg/kg, SD,q8hr (SC,IV)
Bisacodyl	5mg/dog,q8h PO			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Bismuth subsalicylate	1-3ml/kg/day (PO)		60-90ml,TD, q6-12hr (PO)	0.5ml/kg,q4-6hr (PO)
Bleomycin	10 U/m^2 (IV SC) for 3 days, then 10U/ m^2 weekly (maximum cumulative dose 200 U/ m^2			
Bromhexine hydrochloride			0.2-0.5 mg/kg, q24h (IM,PO)	0.1-0.25 mg/kg. q24h (IM,PO)
Bromide	30-40mg/kg, q24h (PO); If administered without phenobarbital higher doses upto 40-50mg/kg may be needed	30-40mg/kg, q24h (PO)		20-40mg/kg, q24hr (PO)
Budesonide	0.125mg/kg, q6-8h (PO); dose interval may increases to q12h when condition improves			
Buserelin			0.02mg/kg, SD (IM,IV,SC)	0.04mg/kg, SD (IM,IV,SC)
Bunamidine hydrochloride	20-50mg/kg (PO)			0.004-0.006mg/kg, SD (IV)
Bupivacaine	1 ml of 0.5% solution per 10 cm for an epidural			
Buprenorphine	0.006-0.02 mg/kg, q4-8h (IV,IM,SC)	0.005-0.01 mg/kg, q 4- 8 h (IV,IM) 0.01-0.02mg/kg, q12h Buccal administration		
Buspirone	2.5-10 mg/dog, q12-24h (PO) 1 mg/kg, q12h (PO)	2.5-5 mg/cat (may be increased to 5-7.5 mg/cat, q24h (PO)		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Busulfan	3-4mg/ m^2, q24h (PO)			
Butorphanol	Antitussive: 0.055 mg/kg, q6-12h (SC) or 0.55 mg/kg (PO) Preanesthetic: 0.2-0.4 mg/kg (with acepromazine) (IV,IM,SC) ANALGESIC:0.2-0.4mg/kg q2-4 h or 0.55-1.1 mg/kg (PO) q6-12h	Analgesic: 0.2-0.8 mg/kg q 2-6 h (IV,SC) 1.5 mg/kg, q4-8h (PO)	0.02-0.04mg/kg, SD (IV,IM)	0.02-0.1mg/kg, SD, q3-4hr (IV,IM)
Calcitriol	2.5 ng/kg, SD	0.25 microgram/ cat, every other day or 0.01-0.04 microgram/kg/ day 2.5 ng/kg for hyperparathyroidism, once daily (PO)		
Calcium carbonate	For phosphate binders: 60-100 mg/kg/day in divided doses (PO) For calcium supplementation: 70-180 mg/kg/day added to food			
Calcium chloride	0.1-0.3 ml/kg IV slowly			
Calcium citrate	20 mg/kg/day added to food	10-30 mg/kg (with meals),q8h (PO)		
Calcium disodium EDTA	25 mg/kg (SC,IM,IV) q6h for 2-5 days			35mg/kg,q12hr IV slow
Calcium gluconate	0.5-1.5 ml/kg IV slowly		150-250mg/kg, SD (slow IV) (SC,IP)	150-250mg/kg, SD (IV slow)
Calcium lactate	0.5 gm/dog/day, divided doses (PO)	0.2-0.5 gm/ cat/day, divided doses (PO)		
Captopril	0.5-2mg/kg, q8-12 h (PO)	3.12-6.25 mg/cat,q8h (PO)		
Carbenicillin	40-50 mg/kg upto 100 mg/kg,q 6-8hr (IV,IM,SC)			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Carbenicillin sodium	10mg/kg,q8h (PO)			50-100mg/kg, q6-12hr (IV)
Carbimazole		5 mg/cat (q8) induction followed by 5 mg/cat, q12h (PO)		
Carboplastin	300 mg/m^2,q3-4 wk IV	200 mg/m^2, q4 Week (IV)		
Carprofen	2.2 mg/kg,q 12h (PO) 4.4 mg/kg, SD PO 2.2 mg/kg, q12h (SC) 4.4 mg/kg, SD (SC)	Doses not available	1.4mg/kg, q24h (IV, SC)	0.7mg/kg, q24hr (IV)
Carvedilol	0.2-0.4 mg/kg titrate upto 1.5 mg/kg if needed,q12h PO			
Cascara sagrada	1-5 mg/kg/day PO	1-2mg/cat/day		
Castor oil	8- 30 ml, q24h PO	4- 10 ml , q24h PO		
Cefadroxil	22-30 mg/kg,q12h PO	22mg/ kg , q24h PO		
Cefamandole				10-30mg/kg, q4-8hr IM,IV
Cefazolin Sodium	20- 35mg/kg, q8hrs (IM,IV) For perisurgical use 22mg/ kg during surgery, q2h			25mg/kg,q6-8hr (IV, IM) 15-20 mg/kg, q8-12h (IV)
Cefixime	10mg/ kg, q12h For cystitis : 5mg/ kg,q12-24 h PO			
Cefoperazone sodium			250 TD, SD (IMM)	30-50 mg/kg, q6-8h (IV,IM)
Ceftriaxone sodium				25-50mg/kg, q12hr IV,IM
Cefotaxime	50mg/kg,q12h IV, IM,SC	20-80mg/kg, q6hrs IV,IM		20-30mg/kg, q6-8hr IV
Cefovecine	8mg/kg, q14 days SC	8mg/kg, q14days SC		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Cefoxitin sodium	30mg/ kg ,q 6-8 h IV			20 mg/kg, q4-6h IV
Cefpodoxine proxetil	5-10mg/kg,q 24 h PO	Dose not established		10-12mg/kg,q8-12hr PO
Ceftazidime	30mg/ kg,q 6 h IV, IM			
Ceftiofur sodium	2.2- 4.4 mg/kg for urinray tract infections,q 24 h SC		1.1-2.2mg/kg, q24hr IM,IV	2.2-4.4 mg/kg,q12-24h (IV, IM)
Cefuroxime			250 mg TD, q12h (IMM) Goats: 40 mg/kg, q12h (IM)	
Cephalexin	10- 30mg/ kg, q6-12 h For pyoderma : 22-35 mg/kg ,q 12 h PO			25-33mg/kg,q6h PO
Cephalothin sodium			55 mg/kg, q6h (SC)	10-30 mg/kg, q6h (IM,IV)
Cephapirin sodium			200 TD, q12h (IMM)	20-30 mg/kg q8-12h (IM,IV) 50 mg/kg, q8-12h (PO)
Cephapirin benzathine			300 TD, SD (IMM)	
Charcoal (activated)			1-3 g/kg, q8-12h (PO)	Adults: 750 g, q8-12h (PO)
Cetirizine	5-10mg/dog up to 2mg/kg,q12 h PO	5mg/cat, q24h (PO)		
Chlorambucil	2-6mg/ sq.m or 0.1-0.2 mg/kg ,q24h initially then q48h (PO)	0.1-0.2mg/kg, q24hrs initially then 48hrs PO		
Chloral hydrate				20-200mg/kg,SD IV 40-100 mg/kg, q6-12h (PO)
Chloramphenicol	40-50mg/kg,q 8 hrs (PO)	12.5 -20mg/kg, q12hrs PO	Not recommended	25-50mg/kg, q6-8hr PO

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Chlorothiazide	20-40 mg/kg, q12h (PO IV)			
Chlorpheniramine maleate	4-8mg/dog up to 0.5mg /kg,q12 h PO	2mg/kg, q12hrs PO		20-60mg/kg, q6-8hr IV,IM
Chlorpromazine	0.5mg/kg, q6-8hrs (IM SC) (cancer chemotherapy-2 mg/kg SC), q3h	0.2 – 0.4mg/kg, q6-8h (IM,SC)	Cattle: 0.22-1.0mg/kg,SD IM Sheep & goats: 0.6-4.4 mg/kg SD IM	Not recommended
Chlortetracycline			6-10 mg/kg, q24h (IM,IV) 10-20 mg/kg, q24h (PO)	
Chorionic gonadotropin	22U/kg IM q24-48h 44U IM once	250 U/cat IM once	Cattle: 2500-5000IU, TD, SD (IV) Cattle: 10,000IU, TD, SD (IM) 250-1000IU, TD, SD (IV,IM)	1000-3000IU,TD,SD (IV,IM,SC)
Cimetidine	10mg/ kg, q6-8 h (IV,IM,PO) In renal failure : 2.5-5mg/kg, q12 h (IV,PO)		8-16mg/kg,q8hr (IV) 50-100 mg/kg, q8h (PO)	6.6mg/kg,q4-6hr (IV) 18 mg/kg, q8h (PO)
Ciprofloxacin	25-30mg/kg (PO) 15mg /kg, q24h Slow IV	10mg/kg, q12hrs PO, 10mg/kg slow IV		
Cisapride	0.1-0.5mg/kg (PO) q8-12h; high dose-0.5-1mg/kg	2.5- 5mg/cat, q8-12 h (PO); high dose 1mg/kg, q8h	Not recommended	0.1mg/kg,q8-12hr (IV) 0.5-1.0 mg/kg, q8-12h (PO)
Cisplatin	60-70mg/sq.m (Administer fluids for diuresis for therapy), q3-4 wks	Do not use		
Clemastine	0.05-0.1mg/kg, q12 hrs PO			
Clindamycin	11-33mg/kg, q12 hrs,(PO) For oral & soft tissue infection: 5.5-33mg/kg, q12 hrs(PO)	11-33 mg/kg for skin &anaerobic infection, q24 hrs (PO) Toxoplasmosis:12.5-25mg/kg q12 hrs for 4 wks(PO)		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Clofazimine		1mg/ kg maximum of 4mg/kg/ day (PO)		
Clomipramine	1-2 mg/kg max of 3mg/ kg, q12 hr (PO)	1-5 mg/ cat, q12-24 hrs (PO)		
Clonazepam	0.5mg/kg, q8-12 hrs,(PO)	0.1-0.2 mg/kg, q12- 24 hrs,(PO)		
Cloprostenol sodium			Cattle: 0.5mg,TD,SD (IM) Sheep & goats: 0.06-0.13 TD, SD (IM)	0.1mg/kg,SD,TD,(IM)
Clopidogril	Give oral loading dose of 2-4mg/kg followed by 1mg/kg, q24hrs (PO)	9mg/ cat (1/4 tablet) q24 hrs (PO)		
Clorazepate	2mg/kg, q12hrs,(PO)	0.2-0.4mg/kg up to 0.5 -2mg/ kg, q12-24 hrs (PO)		
Cloxacillin	20-40 mg/kg q8h,(PO)		Cloxacillin benzathine: 500mg,TD,SD,(IMM) Cloxacillin sodium: 200mg,TD, q12h (IMM)	10-30 mg/kg, q6h (IM,IV)
Colistin				2500IU, q6hr (Slow IV)
Codeine	Analgesic: 0.5-1 mg/kg Antitussive : 0.1-0.3 mg/kg(q4-6 hrs) (PO)			
Colchicine	0.01-0.03 mg/kg, q24 hr (PO)			
Colony stimulating factor	Leukine : 0.25mg/sq.m, q12 hrs,SC/IV Neupogen :0.005 mg/kg(5μg/kg), q24 hrs,SC for 2 week			0.005mg/kg,q24hr IV

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Corticotropin (ACTH)	Response test: Collect pre ACTH sample & inject 2.2 IU/ kg (IM). Collect post ACTH sample in 2 hrs	Response test: Collect pre ACTH sample & inject 2.2 IU/ kg (IM). Collect post ACTH sample in 1-2 hrs		
Cosyntropin	Response test: Collect pre ACTH sample & inject 5μg /kg (IV,IM) and collect sample at 30 and 60 min	0.125 mg (IV IM) and collect sample at 30 min and 60 min after IM administration		
Cyanocobalamin (Vitamin B12)	100-200 microgram/day PO	50-100 microgram/day (PO)		
Cyclophosphamide	Anticancer:50mg/square meter (or), Once daily 4days/wk (PO) or 150-300mg/square meter IV, Repeat in 21 days Immunosuppressive therapy: 50mg/m^2 (Approximately 2.2 mg/kg) (PO) q24h or 2.2 mg/kg once daily for 4 days/wk	6.25-12.5mg/cat, Once daily 4days/wk		2.0mg/kg, q3weeks (IV)
Cyclosporine (Cyclosporine A)	3-7mg/kg/day for atopic dermatitis	5mg/kg, q24h (PO) (or) 7.5mg/kg, PO		
Cyproheptadine	Antihistamine:1.1mg/kg, q8-12h (PO)	Appetite stimulant:2mg/cat (PO)		0.25-1.2mg/kg,q12-24hr,PO
Cytarabine (Cytosine arabinoside)	Lymphoma:100mg/square meter once daily (or)50mg/square meter twice daily for 4days(IV,SC) Meningoencephalitis:200mg/square meter, Infused over 8 hours, repeated twice in 48hours,IV	100mg/square meter , Once daily for 2days		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Dacarbazine	200mg/square meter for 5 days (IV); 800-1000 mg/square meter (IV) (q3wk)			
Dalteparin	100-150 units/kg, q8h (SC)	180 units/kg , q6h,(SC)		50IU,q12hr (SC)
Danazol	5-10mg/kg, q12h(PO)			
Danofloxacin			8 mg/kg, SD(SC) 6 mg/kg, q48h (SC Only repeat once)	
Dantrolene	For prevention of malignant hyperthermia: 2-3mg/kg (IV) For muscle relaxation:1-5mg/kg(q8h) (PO)	0.5-2mg/kg, q12h,(PO)		2 mg/kg, q6hr (IV) 2-10 mg/kg, q24h (PO)
Dapsone	1.1mg/kg , q8-12h,(PO)	Do not use		
Darbazine (prochlorperazine +isopropamide)	0.14-0.2ml/kg (SC) Dog 2-7kg:1-#1 capsule (PO) Dog 7-14 kg:1-#2 capsule (PO) Dog more than 14 kg:1-#3 capsule, q12h (PO)	0.14-0.2ml/kg,q12h (SC)		
Desferoxamine	10mg/kg q2h for two doses,(IV,IM) then 10mg/kg q8h for 24 hrs		10 mg/kg,SD (IM,IV)	10mg/kg,SD (IM,IV)
Detomidine hydrochloride			0.002-0.02 mg/kg,SD (IV,IM)	0.005-0.08mg/kg,SD, q2-4hr (IV,IM)
Deracoxib	3-4 mg/kg for 7 days (PO) q24h or 1-2 mg/kg q24h (PO) for long term use	1mg/kg (PO) One time dose		
Desmopressin acetate	Diabetes insipidus:2-4 drops (2µg), q12-24h (Intranasally or In eye) Animal oral dose:0.05-0.1mg/dog then increase to 0.1-0.2mg/dog			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
	q12h (PO) Von Willebrand's disease treatment: 1 microgram/kg,(SC, IV) diluted in 20 ml saline administered over 10 min			
Desoxycorticosterone pivalate	1.5-2.2 mg/kg , q25 days,(IM)			
Dexamethasone	Antiinflammatory: 0.07-0.15 mg/kg (IV IM PO) q12-24h Dexamethasone suppression test: 0.01mg/kg (IV); collect sample at 0,4 and 8 hr	Dexamethasone suppression test: 0.1mg/kg (IV); collect sample at 0,4 and 8 hr	Cattle: 20-30TD (Induction of parturition) SD (IM) Cattle: 0.02-2 mg/kg (antinflammatory), q24h (IV,IM) Cattle: 5-20 TD (ketosis) SD q24h (IM)	0.01-0.2mg/kg,q24hr (IV,IM,PO) (anti-inflammatory) 0.5-2 mg/kg, SD (IV,IM)
Dexmedetomidine	Sedative and analgesic 375 mg /square meter,(IV) 500 mg/square meter,(IM) Preanesthetic : 125 mg/ square meter,(IM)	Sedative and analgesic :40 micro gram /kg,(IM)		
Dimercaprol(BAL)	4 mg/kg, q4h (IM)	4 mg/kg, q4h,(IM)		
Dextran	10-20 ml/kg (IV) to effect			
Dextromethorphan	0.5-2 mg/kg, q6-8 h,(PO)			
Dextrose solution 5%	40-50 ml/kg, q24 h,(IV)			
Diazepam	Preanesthetic: 0.5 mg/kg (IV) Status epilepticus: 0.5 mg/kg (IV); 1 mg/kg rectal (repeat if necessary)	Appetite stimulant:0.2 mg/kg, (IV)	0.4mg/kg,SD,(IV)	0.05-0.4 mg/kg SD (IV) 0.5 mg/kg, SD (IM)
Dioctyl calcium sulfosuccinate	50-100 mg/dog, q12-24h,(PO)	50mg//cat , q12-24 h,(PO)		
Dichlorvos	26.4 -33 mg/kg,(PO)	11 mg/kg, (PO)		35mg/kg,SD,(PO)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Dicloxacillin	25 mg/kg, q6h (IM) oral doses not absorbed			10mg/kg,q6hr, (IM)
Diethylcarbamazine	Heartworm prophylaxis :6.6 mg/kg, q24h,(PO)		22 mg/kg, q24h (IM)	
Dioctyl sodium sulfosuccinate	50- 200 mg/dog , q8-12h,(PO)	50mg/cat , Q12-24h,(PO)		
Diethylstilbestrol	0.1-1 mg/dog, q24 h,(PO)	0.05-0.1 mg/cat, q24h (PO)		
Digoxin	Less than 20 kg BW: 0.01 mg/kg, q12h,(PO) More than 20 kg: 0.22mg/ square meter, q12 h,(PO) Rapid digitalization: 0.0055-0.011mg/kg, q1 h,(IV) to effect	0.008-0.01 mg/kg, q48h (PO)	0.022 loading, 0.0034,q4hr (IV)	0.002mg/kg,q12h (IV) 0.01-0.02 mg/kg, q12-24h (PO)
Dihydrostreptomycin			11mg/kg,q12hr (IM,SC)	11mg/kg,q12h (IM,SC)
Dihydrotachysterol (Vitamin D)	0.01 mg/kg/day(PO) For acute treatment 0.02 mg/kg initially then 0.01-0.02 mg/kg, q24-48 h(PO)			
Diltiazem	0.5-1.5 mg/kg, q8h,(PO) 0.25 mg/kg over 2 mins(IV)	1.75-2.4 mg /kg q8h For Dilacor XR or cardizem CD :10 mg/kg (PO) once daily		
Dimethyl glycine				1-2mg/kg,q24hr (PO)
Dimethyl sulfoxide			Not recommended	0.5-2mg/kg,q12-24hr (IV as 10% solution slowly, PO, Topical) 100g, TD, q12-24h

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Dinoprost tromethamine			Cattle: 25 TD (estrus induction), 10-12 days (IM) Cattle: 25 TD (abortifacient), SD (IM) Ewe: 8 TD (estrus induction), days 5 and 11 of cycle (IM) Doe: 8 TD (estrus induction) days 4 and 11 of cycle (IM) Ewe: 10-15 TD (abortifacient)(IM) Doe: 5-10 TD (abortifacient)(IM)	0.002-0.01 mg/kg, SD (IM)
Dimenhydrinate	4-8mg/kg, q8h,(PO,IM,IV)	12.5 mg/cat q8h (IM IV PO)		
Dioctyl sodium sulfosuccinate				10-20 mg/kg, q48h (limit 2 doses) (PO)
Diphenhydramine	25-50mg/dog q8h(IV,IM,PO)	2-4mg/kg,q 6-8 hr 1mg/kg,q 6-8 hr	0.5-1 mg/kg, q6-8h (IV,IM)	0.5-1 mg/kg, q6-8h (IV,IM)
Diprenorphine			Cattle: 0.03 mg/kg, SD (IV) Sheep: 0.015 mg/kg, SD (IV)	Horses: 0.03 mg/kg, SD (IV) Donkeys: 0.015 mg/kg, SD (IV)
Dipyrone			50 mg/kg, SD (IM,IV,SC)	11-22 mg/kg, SD (IV, IM)
Diphenoxylate	0.1-0.2mg/kg q8-12h(PO)	0.05-0.1mg/kg q12h (PO)		
Diphenyl hydantoin	Antiepileptic:20-35mg/kg q8h(PO) Antiarrhythmic:30mg/kg (PO) q8h or 10mg/kg (IV) over 5 min			
Diphosphonate disodium etidronate	5mg/kg/day(PO)	10mg/kg/day (PO)		
Dipyridamole	4-10mg/kg q24h(PO)			
Dirlotapide	Start with 0.01mg/kag/day not exceed 0.02mg/kday(PO)	Not administer		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Disopyramide	6-15mg/kg q8h(PO)			
Dithiazanine iodide	Heartworm: 6.6-11mg/kg(PO) q24h for 7-10 days Other parasites:22mg/kg (PO)			
Divalproex sodium	60-200mg/kg(PO) q8h or 25-105mg/kg/day when administered with phenobarbitol q8h (PO)			
Dobutamine	5-20µg/kg/min(IV)	2 µg /kg/min IV infusion		1-10 µg /kg/min IV infusion
Docusate calcium	50-100mg/dog q12-24h(PO)	50mg/cat q12-24h (PO)		
Docusate sodium	50-100mg/dog q8-12h(PO)	50mg/cat q12-24h (PO)		
Dolasetron mesylate	For preventing nausea,vomition 0.6mg/kg(q24 h)(IV,PO) Treating nausea,vomition-1mg/kg(IV,PO) SD			
Domperidone	2-5mg/animal(PO)			
Dopamine	2-10 µg /kg/min IV infusion (IV)	2-10 µg /kg/min IV infusion (IV)	2-10 µg /kg/min IV infusion (IV)	1-10 µg /kg/min IV infusion (IV)
Doramectin	0.6 mg/kg SC for demodicosis		0.2 mg/kg,SD (IM,SC)	
Doxapram	5-10mg/kg (IV) Neonate:1-5mg(IV,SC,sublingual umblical vein)		5-10 mg/kg,SD (IV)	0.02-1 mg/kg,SD (IV)
Doxorubicin	30mg/m^2 q21days (IV) 20kg use 30mg/m^2 20kg use1mg/kg	20mg/ m^2 q3wk(IV) or approximately 1-1.25 mg/kg (IV) q3wk		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Doxycycline	3-5mg/kg,q12h,(PO, IV) 10mg/kg,q24h,(PO) For rickettsia: 5mg/kg,q12h	3-5mg/kg,q12h(PO, IV) 10mg/kg,q24h(PO)		10 mg/kg,q12 (PO, don't use IV)
Doxycycline succinate			0.5 mg/kg, q12h (PO,IM,SC)	0.55 mg/kg, q8h (IM,SC,Slow IV)
Edrophonium	0.11-0.22mg/kg(IV)	0.25-0.5mg/cat(IV) total dose	0.5-1 mg/kg, SD (IV)	0.1-0.5 mg/kg ,SD (slow IV)
Enalapril	0.5mg/kg q12-24h(PO)	0.25-0.5mg/kg q12 -24h(PO)		
Enflurane	Induction:2%-3% Maintenance:1.5%-3%			
Enilconazole	Nasal aspergillosis : 10mg/kg q12h (instilled into nasal sinus for 14 days) 10% solution diluted 50/50 with water Dermatophytes: 10% solution to 0.2%and wash lesion with solution four times at 3 to 4 days interval			
Enoxaparin	0.8mg/kg, q6 hr(SC)	1.25 mg/ kg q6 hr (SC)		
Enrofloxacin	5-20 mg/kg/day (PO,IM)	5 mg/kg (PO) (Do not exceed dose)	7.5-12.5 mg/kg,SD(SC) 2.5-5 mg/kg, q24h (SC) for 3-5 days)	5 mg/kg, q12h (PO) 7.5 mg/kg, q24h (PO)
Ephedrine	Vasopressor: 0.75 mg/kg Repeat as needed(IM,SC)			0.7 mg /kg, q12h (PO)
Epinephrine	Cardiac aresst: 10-20µg/kg or 200 µg /kg intratracheal (may be diluted in saline before administration) Anaphylactic shock: 2.5-5 µg /kg IV or 50 µg /kg intratracheal (may be diluted in saline)		0.01-0.02 ml/kg, SD (IM,SC) 0.1-0.2 ml/kg,SD,(IM, SC)	0.01-0.02 ml/kg, SD (IM,SC) 0.1-0.2 ml/kg,SD,(IV)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Eprinomectin			0.5 mg/kg SD Topical 1 mg/kg SD (SC)	
Epoetin alpha (erythropoietin)	35 to 50 U/kg three times per week to 400 U/kg/wk (IV,SC) (Adjust dose to haematocrit of 0.30-0.34)	Start with 100 units /kg three times per week (Adjust dose based on haematocrit)		
Epsiprantel	5.5 mg/kg(PO)	2.75 mg/kg(PO)		
Ergocalciferol (Vitamin D2)	500-2000 units/kg q24 hr/day (PO)			
Erythromycin	Antibacterial dose:10-20 mg/kg, q 8-12 hr (PO) Prokinetic dose:0.5-1 mg/kg, q8 hr (PO)		2.2 -15 mg/kg, q12-24h (IM)	0.1 mg/kg (for ileus),infusion per hour (IV)
Erythromycin estolate, ethyl succinate			300 TD 600 TD (dry cows) q12h (IMM)	25-37.5 mg/kg, q6-12h (PO)
Esmolol	500 mcg/kg (IV) which may be given as 0.05-0.1 mg/kg slowly every five minutes or 50 – 200 mcg/kg/min infusion			
Estradiol cypionate (ECP)	22-44 mcg/kg (total dose not exceed 1mg) (IM)	250 mcg/cat(IM) Between 40 hrs and 5 days of mating	-	5-10 TD (estrus induction),SD (IM)
Estrone sulfate				0.04 mg/kg, q12hr (IM)
Etidronate disodium	5 mg/kg q24 hr (PO)	10 mg/kg q24 hr (PO)		
Etodolac	10 -15 mg/kg q24 hr (PO)	Dose not established		
Etretinate	1 mg/kg with food. less than 15 kg: 10mg per dog (q24 h) (PO) More than 15 kg: 10 mg per dog (PO)q12 hr	2 mg/kg q24 hr		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Famotidine	0.1-0.2 mg/kg, q12 h (PO, IM, SC, IV) 0.5 mg/kg(Q 24 hr) (IM, SC, PO, IV)	0.2 – 0.25 mg/kg, q12 -24 hr (IM, IV, SC, PO)		1.9-2.8 mg/kg, q8-12h (PO) 0.2-0.4mg/kg,q8-12h (IV)
Febantal			5-10mg/kg, SD (PO)	6mg/kg, SD (PO)
Felbamate	Start with 15 mg/kg and increase gradually to maximum of 65 mg/kg q 8 hr (PO)			
Fenbendazole	50 mg/kg/day For three days (PO)		5-10mg/kg,SD (PO)	Adults: 5mg/kg,SD (PO) Foals: 10mg/kg,SD (PO)
Fenoterol				2-4mg/kg,q6-12hr (IH)
Fenprostalene			0.002mg/kg,SD (SC)	0.001mg/kg,SD (IM)
Fentanyl	0.02-0.04mg/kg q2h 0.01mg/kg(with acetylpromazine or diazepam) For analgesia :0.01mg/kg (IM, IV, SC)			
Fentanyl transdermal	10-20kg:50µg/hr Patch q72hrs	25mcg Patch every 118 hr		2x20mgpatch per 400kg
Ferrous sulphate	100-300mg/dog q24hrs (PO)	50-100mg/cat q24hrs (PO)	10-30mg/kg,q24hr (PO)	10-20mg/kg, q24hr (PO)
Finasteride	BPH:5mg tablet/dog q24hrs (PO)			
Firocoxib	5mg/kg q24hrs (PO)	1.5mg/kg once long -term safety in cats has not been determined		
Florfenicol	20mg/kg q6hrs (PO, IM)	22mg/kg q12hrs (IM, PO)	20mg/kg,q48hr (IM) repeat once 40mg/kg, SD (IM)	

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Fluconasole	10-12mg/kg ,q24 h For malassezia - 5mg/kg, q12hrs (PO)	50mg/cat q12-24hrs (PO)		4mg/kg,24hr (PO)
Flucytosine	Do not use	25-50mg/kg (PO) q6-8h (up to a maximum dose of 100mg/kg) (PO) q12hrs		
Fludrocortisone	0.2-0.8mg/dog or 0.02mg/kg(13-23 mcg /kg) q24hrs (PO)	0.1-0.2mg/cat q24hrs (PO)		
Fluprednisolone acetate				0.01-0.04mg/kg, SD (IM)
Fluprostenol				0.55μg/kg, SD (IM)
Flumazenil	0.2mg (total dose) (IV) as needed			0.01-0.02mg/kg, SD (Slow IV)
Fluticasone				2-4μg/kg, q6-12hr (IH)
Flumethasone	Antiinflammatory:0.15-0.3mg/kg, q12-24hrs (IV, IM, SC)	Antiinflammatory:0.15-0.3mg/kg q12-24hrs		0.002-0.008 mg/kg,SD (IM, IV, IA)
Flunixin meglumine	1.1mg/kg (q24h) (IV, IM, SC) Or 1.1mg/kg per day, 3 days /week (PO) Opthalmic - 0.5mg/kg (IV) once		1.1-2.2mg/kg, q12-24hr (IM, IV)	0.25-1.1 mg/kg, q6-24hr (IV, IM, PO)
5-fluorouracil	150mg/sq.m, Once per week (IV)	Do not use		
Fluoxetine	1-2mg /kg/day (PO)	0.5-4mg/cat q24hrs (PO)		
Follicle - stimulating hormone(FSM)	75 U/day (IM) for 7 days		5 TD, q12hr (IM, SC)	10-50 TD,SD (IV, IM, SC)
Fomepizole	20mg/kg initially IV then 15 mg/kg @12 and 24hrs interval then 5mg//kg @36 hrs interval (repeat q12 hr if necessary)			
Folic acid				40-75 mg TD,SD (PO)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Folinic acid				50-100 TD,SD, q24hr (PO)
Furazolidone	4mg/kg (PO) q12 hr for 7 to 10 days		Calves: 10mg/kg, q24hr (PO)	4mg/kg, q8hr (PO)
Furosemide	2 -6mg/kg q8-12hr (or as needed) (IV, IM,SC, PO)	1-4mg/kg q8-24 hr (PO, IV, IM, SC)	Adult cattle: 0.5-1mg/kg, q12-24hr (IM)	0.25-3 mg/kg,SD (IV, IM)
Framycetin sulfate			5mg/kg, q12 hr (IM) Calves: 10mg/kg, q24hr (PO)	
Gabapentin	Anticonvulsant:2.5-10mg/kg, q8-12 h (PO) For analgesia: 10-15mg/kg, q8 hr (PO)	Anticonvulsant dose: 2.5-10mg/kg, Q8-12 hr For analgesia:10-15mg/kg, q8hr (PO)		
Gallamine triethiodide			Cattle: 0.5mg/kg then 0.1mg/kg to effect,SD (IV) Sheep: 0.4 mg/kg,SD (IV)	1mg/kg then increments of 0.2mg/kg,SD (IV)
Gemfibrozil	7.5mg/kg q12 hr (PO)			
Gentamicin	9-14mg/kg q24 hr (IV, IM, SC)	5-8mg/kg q24 hr (IV, IM, SC)	2.2-6.6mg/kg, q12-24hr (IM) 100-150 TD, q12hr (IMM)	2.2mg/kg, q8hr (IV, IM) 6.6 mg/kg, q24hr (IV, IM)
Glipizide	Not recommended	2.5-7.5mg/cat (PO) q12h,usual dose is 2.5mg /cat initially,then increase to 5mg/cat q12 hr		
Glauber's salts			1-2mg/kg,SD (PO)	1-2mg/kg, SD, q2 4hr (PO)
Glucosamine chondroitin sulfate	1-2 RS (regular strength) capsules/day (2-4 capsules of double strength for large dogs)	1 regular strength capsule/day (PO)		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Glyburide	0.2 mg/kg q24hrs (PO)	0.625 mg/cat		
Glycerin	1-2 ml/kg 8hrs (PO)			
Glycerol			Cattle: 180 mL TD, q12hr (PO) Sheep: 90mL TD, q12hr (PO)	
Glycerol guaiacolate ether				110mg/kg, SD (IV)
Glycopyrrolate	0.005-0.01 mg/kg (IV, IM, SC)			0.001-0.01 mg/kg,SD, q12-24hr (IV, IM)
Glycosaminoglycan (polysulfated)				250 TD,SD, 7days,(IA) 1mg/kg, 5days(IM)
Gold sodium thiomalate	1-5 mg IM on first week, then 2-10 mg IM on second week, then 1 mg/kg once/week IM maintenance			
Gonadorelin (GnRH, LHRH)	50-100 mcg/dog/day(I/M) 24-48 hrs	25 mcg/cat Once(I/M)	100µg TD,SD(IM)	
Granisetron	0.01 mg/kg (10 mcg/kg) (I/V)			
Griseofulvin (microsize)	50 mg/kg (Upto a maximum dose of 110-132 mg/kg/day in divided treatments q24 hrs(PO)			
Griseofulvin (ultramicro size)	30 mg/kg/ day in divided treatments(PO)			
Griseofulvin			10-20mg/kg, q24hr(PO)	5-10mg/kg, q24hr(PO)
Growth hormone	0.1 U/kg three times per week for 4-6 weeks (Usual human pediatric dose is 0.18-0.3 mg /kg /wk) (SC, IM)			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Guaifenesin	Expectorant:3-5 mg/kg (PO) q8h Anaesthetic adjunct: 2.2 ml/kg/hr of a 5% solution q8 hrs(IV)	Expectorant:3-5 mg/kg (PO) q8 hrs	66-130mg/kg, SD(IV)	110mg/kg,give first one third to cause recumbency, SD (IV)
Halothane	Induction:3% Maintenance:0.5%-1.5%			
Haloxon				50-70mg/kg, SD(PO)
Hemoglobin glutamer	10-30 ml/kg at a rate not to exceed 10ml/kg/hr(IV) One time dose	3-5 ml/kg One time dose(Slow IV)		
Haemoglobin (bovine, polymerized)				10-30mL/kg, SD (IV)
Heparin sodium	100-200 units/kg IV loading dose; then 100-300 units /kg (q6-8 hrs) (SC) Low dose prophylaxis:70 U/kg (q8-12 hrs))(SC)	Low dose prophylaxis:70U/kg, q8-12 h(SC)		30-100mg/kg,6-12 hr(SC,IV)
Hetastarch	10-20 ml/kg/day(IV)	5-10 ml/kg/day(IV)		10-20mL, SD(Slow IV)
Hyaluronate sodium				10-50 TD,SD(IA)
Hydralazine	0.5 mg/kg (initial dose) titrate to 0.5-2 mg/kg q12 hrs(PO)	2.5 mg/cat q12-24 hrs(PO)		0.5-1.5mg/kg, q12 hr (PO)
Hydrochlorothiazide	2-4 mg/kg, q12 hrs(PO)		0.25-0.5mg/kg, q12-24hr(IV,IM)	0.5mg/kg, q24hr(PO)
Hydrocodone bitartrate	0.22 mg/kg q4-8 hrs(PO)	No dose available		
Hydrocortisone	Replacement therapy 1-2mg/kg, q12h(PO) Anti-inflammatory 2.5-5mg/kg, (PO)q12h		0.1-1mg/kg, q24h, SD (IV,IM)	
Hydrocortisone sodium succinate	Shock 50-150mg/kg (IV) Anti-inflammatory 5mg/kg(IV), q12h			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Hydroxyethylstarch	10-20ml/kg (IV to effect)			
Hydromorphone	0.22 mg/kg (IM SC). Repeat every 4-6h or as needed for pain treatment	0.1-0.2mg/kg (IM SC) q2-6h or as needed or 0.05-0.1 mg/kg IV q2-6h		
Hydroxyurea	50mg/kg, Once daily, 3 days/week (PO)	25 mg/kg, Once daily, 3 days /week (PO)		
Hydroxyzine	2mg/kg, q12h(PO,IV,IM)	Safe dose not established		0.5-1mg/kg, q12hr (IM,PO)
Hyoscine				0.3mg/kg, SD (Slow IV)
Imipenem	3-10mg/kg, q6-8h(IV,SC or IM)			15-20 mg/kg ,q4-6 hr(Slow IV)
Imidocarb dipropionate	2 mg/kg (IM SC) once for premunition; 6.6 mg/kg (IM SC) repeat in 14 days for sterilization		1.2 mg/kg SD(SC)	2-4 mg/kg,q24h (IM) 4.4mg/kg, q72h (IM) (Total of 4 treatments)
Imipramine	2-4 mg/kg, q12-24h(PO)	0.5-1mg/kg, q12-24(PO)		0.55-1.5 mg/kg ,q8hr (IM,IV,PO)
Insulin, NPH isophane, ultralente Or PZI	Start with 0.75 U/kg,q 12h(SC)	0.5 to 1.0 U/kg, Usually twice /day(SC)		
Insulin, protamine zinc suspension			200 IU, TD (SC)	0.15 IU/kg q12h (IM,SC)
Interferon alfa-2a human recombinant	2.5 million U/kg (IV) once daily for 3 days	1 million U/kg (IV) once daily for 5 consecutive days @0,14,60 days		0.1-0.3 IU, q24 hr (PO)
Isotretinoin	1-3 mg/kg/day (upto maximum recommended dose of 3-4 mg/kg/day) (PO)	Dose not established		
Iodide sodium			66 mg/kg, SD (IM) (Repeat once in 7 days)	20-40mg/kg, q24hr (IV, PO)
Iodochlorhydroxyquin				20 mg/kg, q24 hr (PO)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Ipecac syrup	3-6ml/ dog (PO)	2-6ml/cat (PO)		
Ipodate	15 mg/kg,q12h(PO)	50mg/cat q12h		
Iron dextran			2mg/kg,SD(IM)	
Iron cacodylate				2mg/kg, SD (IV)
Isoflurane	Induction:5%Maintenance:1.5%-2.5%			
Isofluperdone acetate			10-20 mg TD, SD (IM)	5-20 mg TD,SD (IM)
Isoniazid			11-25mg/kg, q24hr (PO)	5-20 mg/kg, q24 h (PO)
Isoproterenol	10 µg/kg, q6h(IM,SC)			
Isosorbide dinitrate	2.5-5 mg/animal, q12h (PO) or 0.22-1.1 mg/kg (PO) q12h			
Isosorbide mononitrate	5 mg/dog (PO) 2 dose/day 7 hrs apart			
Isoxsuprine hydrochloride				0.4-1.2 mg/kg, q8-12h (PO)
Itraconazole	2.5mg/kg, q12h(PO) 5mg/kg q24h (PO) For dermatophytes-3mg/kg/day for 15days q24h(PO) Malassezia-5mg/kg q24h for 2 days repeated each week for 3 week (PO)	1.5-3mg/kg upto 5mg/kg (PO) for 15 days (PO)		3-5mg/kg,q12-24hr (PO)
Ivermectin	Heartworm preventive: 6µg/kg(PO) q30 days Microfilaricide:50 µg /kg 2 weeks after adulticide therapy (PO) Ectoparasite therapy-200-300mcg/kg (IM,PO,SC) Endoparasites-200-400 µg /kg, Weekly (PO,SC)	Heartworm preventative:24 µg /kg Q30 days (PO) Ectoparasite therapy-200-300µg/kg (IM,SC,PO) Endoparasites-200-400µg /kg, Weekly (SC,PO)	0.2 mg/kg,SD (SC, PO)	0.2 mg/kg,SD,(PO)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
	Demodex : start with 100 µg /kg then increased to 600 µg /kg/day for 60to 120 days (PO)			
Kanamycin	10mg/kg (q6-8h) or 20mg/kg (q24h) (IM,SC,IV)			
Kaopectate (Kaolin +pectin)	1-2ml/kg, q2-6h,(PO)		0.25-1 ml/kg,q4hr (PO)	2-4 ml/kg, q8-12hr (PO)
Ketamine	5.5-22mg/kg(IV,IM)	2-25mg/kg(IV,IM)	2 mg/kg ,SD(IV) 4 mg/kg,SD,(IM)	1.1 mg/kg,SD (IV)
Ketaconazole	10-15mg/kg, q8-12,(PO) Malassezia canis-5mg/kg, q24h(PO) Hyperadrenocorticism-15mg/kg, q12h(PO)	5-10mg/kg, q8-12h (PO)		5-30mg/kg, q12-24hr (PO)
Ketoprofen	1mg/kg for upto 5 days (q24h)(PO)or 2 mg/kg for one dose(IV,IM,SC)	1mg/kg for upto 5 days (q24h)(PO)or 2 mg/kg for one dose(IV,IM,SC)	2-4 mg/kg, q24hr (IM,IV)	2.2 mg /kg, q24hr (IM,IV)
Ketorolac tromethamine	0.5mg/kg for not more than 2 doses,q12h(PO,IV,IM)		Goats: 0..3-0.7 mg/kg, q8hr (IM,IV,SC,PO)	0.5mg/kg, SD (IV)
Lactated Ringer's solution	Maintenance-55-65ml/kg/day (IV) For severe dehydration-50ml/kg/hr (IV) For shock-90ml/kg (IV)	For shock 60-70ml/kg (IV)		
Lactulose	Constipation: 1ml/4.5kg (to effect) q8h (PO) Hepatic encephalopathy-0.5ml/kg, q8h,(PO)	Hepatic encephalopathy:2.5-5ml/cat,q8h (PO)		120-300mg/kg, q12hr (PO)
Lasalocid			1 mg/kg,q 24 hr,(PO)	
Leucovorin (folinic acid)	With methotrexate administration-3mg/sqm (IV,IM,PO) Antidote for pyrimethamine toxicosis -1 mg/kg,q24h (PO)			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Levamisole	Hook worm:5-8mg/kg (PO) once upto 10mg/kg PO for 2 days Microfilaricidae:10 mg/kg q24h for 6-10 days (PO) Immunostimulant:0.5-2mg/kg 3 times/week (PO) For lungworn: 20-40mg/kg q48 h (PO) for 5 treatments	4.4mg/kg once(PO)	5.5-11 mg/kg,SD(PO) 3.3-8 mg/kg,SD(SC)	8-11 mg/kg, q24hr (PO)
Levetiracetam	20mg/kg q8h (PO)	30mg/kg q12hr(PO)		
Levodopa (L-dopa)	6.8mg/kg initially then 1.4mg/kg q 6h (PO)			
Levothyroxine sodium (T4)	18-22 µg /kg q12 hr,(PO)	10-20 µg /kg/day (PO)		0.02 mg/kg , q24hr (PO)
Lidocaine	Antiarrhythmic:2-4mg/kg to a maximum dose of 8mg/kg over 10 mins period (IV); 25-75 µg/kg/min (IV infusion); 6mg/kg q1.5 h (IM) For epidural:4.4mg/kgof 2% solution	0.25-0.75mg/kg (IV) slowly or 10-40 µg/kg/min infusion For epidural: 4.4mg/kg of 2% solution		1.3 mg/kg as bolus then 0.05 mg/kg/min infusion (IV)
Lincomycin	15-25mg/kg q12h (PO) Canine pyoderma: doses as slow as 10mg/kg, q12h	15-25mg/kg q12hr(PO)	5-10 mg/kg,q 12-24 hr(IM)	
Linezolid	10mg/kg q8-12hr(PO,IV)	10mg/kg q8-12hr(PO,IV)		
Liothyronine (T3)	4.4mcg/kg q8hr,(PO)			
Lisinopril	0.5mg/kg q24hr(PO)	No dose established		
Lithium carbonate	10mg/kg(PO) q12hr	Not recommended		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Lomustine	70-90mg/sqm every 4wk(PO) For brain tumour: 60-90mg/sqm q6- 8wk,(PO)	50-60mg/sqm q3-6wk,(PO) 10mg/cat every 3 wk,(PO)		
Loperamide	0.1mg/kg q8-12hr,(PO)	0.08 to 0.16mg/kg q12 hr,(PO)		0.1-0.2 mg/kg,q6hr (PO)
Lufenuron	10 mg /kg q30 days (PO)	30mg/kg q30 days (PO) 10mg/kg (SC) q6months		5-20 mg/kg, q24hr (PO)
Lufenuron+milbemycin oxime	Administer 1 tab q30 days,(PO)			
Luprositol			Cattle: 7.5-15 TD, SD (IM)	7.5mg TD,SD (IM)
L-Lysine		Paste formulation: 1-2ml/cat Adult cat: approx 400mg/cat Kittens:1ml/cat, (PO)		
Magnesium citrate	2-4ml/kg,(PO)			
Magnesium oxide			1000-2000 mg/kg SD (PO)	
Magnesium hydroxide (milk of magnesia) antacid	Antacid:5-10ml/kg, q4-6h Cathartic: 15-50ml/kg (PO)	2-6ml/cat, q24h,(PO)	Cattle: 400-450 g TD,q8-24 hr (PO) Sheep: 10-30 g,TD, q 8-24 hr (PO)	0.5 ml/kg, q8hr (PO)
Magnesium sulfate (Epsom salt)	Dogs:8-25gm/dog, q24hr (for treating arrhythmias) 0.15-0.3mEq/kg slowly IV 5-15min followed by 0.75-1.0 mEq/kg/day; Fluid supplementation: 0.75-1.0 mEq/kg/day (PO)	2-5gm/cat, q24h(PO)	0.1 mg/kg,SD,(SC) 0.02 mg/kg, SD (with calcium gluconate) (Slow IV)	0.2-1 mg/kg , q24 hr (PO) 2.2-6 mg/kg (for ventricular tachycardia),1 min (IV boluses every minute until conversion or total dose of 60 mg/kg) 50mg/kg/h for 1 hour then 25mg/kg/h as CRI (for

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
				presumed neonatal hypoxic encephalopathy)
Mannitol	Diuretic: 1gm/kg of 5-25% solution IV to maintain urine flow. Glaucoma or central nervous system edema: 0.25-2gm/kg of 15-25% solution IV over 30-60 min (repeat in 6hr if necessary) (IV); 2.75-5.55 mg/kg, q24h (PO)		1-3g /kg ,SD (IV)	0.25-2g/kg,SD (Slow IV)
Marbofloxacin		2.75-5.55mg/kg, q24h q24h(PO)	2 mg/kg, q24hr (IM,IV,SC)	
Maropitant	Dog;1mg/kg (SC) once daily for upto 5days. 2mg/kg (PO) once daily for upto 5days. For motion sickness:8mg/kg PO once daily for upto 2days	Cat;1mg/kg once daily (PO,IV, SC)		
MCT oil (medium chain triglycerides)	1-2ml/kg/day in food			
Mebendazole	22mg/kg(with food) q24 h for 3days(PO)			8.8 -20 mg/kg, SD (PO)
Meclizine	25mg , q 24h (for motion sickness,administer 1hr before travelling)(PO)	12.5mg,q24h(PO)		
Meclofenamic acid (meclofenamate sodium)	1mg/kg/day for upto 5days(PO)	Not recommended		2,2 mg/kg q 12-24 hr(PO)
Medroxyprogesterone acetate	1.1-2.2 mg/kg, q7day (IM) For behavioral use 10- 20mg/kg (SC) For prostate 3-5mg/kg (SC,IM) every 3-4weeks			0.3-0.6mg/kg q24h (IM)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Megestrol acetate	Dog: proestrus 2mg/kg PO q24h for 8 days. Anestrus:0.5mg/kg PO q24h for 30 days Behaviour:2-4mg/kg q24h for 8days (reduce dose for maintenance)	Dermatological therapy or urine spraying:2.5-5mg/cat (PO) q24h for 1week,then reduce to 5mg once or twice/wk Suppress estrus: 5mg/cat/day for 3days then 2.55mg once /wk for 10wk		
Melarsomine	Class1- 2dogs: 2.5mg/kg/day for 2 consecutive days. Class 3 dogs: 2.5 mg/kg once, then in 1 month two additional doses 24hr apart.			
Meloxicam	0.2mg/kg initially, then 0.1mg/kg, q24h (PO) Injection:0.1mg/kg (IV,SC)	0.3mg/kg, One time injection(SC) 0.05mg/kg(for chronic use), q24-48hr(PO)	0.5 mg/kg ,SD (IV,SC) 0.5-1 mg/kg, q24-48hr (PO)	0.6 mg/kg, q24 hr(IV,PO)
Melphalan	1.5mg/m2(or 0.1-0.2mg/kg), q24hr for 7-10days(repeat every 3week(PO)			
Meperidine	5-10mg/kg as often as q2-3h (or as needed) (IV,IM)	3-5mg/kg, q2-4hr(or as needed(IV,IM)	3-4 mg/kg,SD (IM,SC)	1-2 mg/kg ,SD (IM) 0.2-0.4 mg/kg, SD(IV)
Mepivacaine	Variable dose for local infiltration For epidural-0.5ml of 2%solution, q30sec until reflexes are absent			
6-Mercaptopurine	50mg/m2, q24hr(PO)	Don’t use		
Meropenem	8.5mg/kg upto 12mg/kg, q12hr (SC); 24mg/kg, q12h (IV)	8.5mg/kg upto 12mg/kg, q12hr (SC); 24mg/kg, q12h (IV)		
Mesalamine	Veterinary dose has not been established the usual human oral dose is 400-500mg, q6-8h			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Mezlocillin				25-75 mg/kg, q6hrs(IV)
Metaproterenol	0.325-0.65mg/kg, q4-6hr(PO)			
Methadone	0.5-2.2mg/kg or 0.5-1mg/kg for analgesia (q3-4hr) (IV,SC,IM)	0.2-0.5mg/kg (SC,IM); 0.05-0.1mg/kg upto 0.2mg/kg for analgesia, q3-4hr(IV)		0.05-0.2 mg/kg,SD(IV,IM)
Methazolamide	2-3mg/kg, q8-12hr(PO)			
Methenamine hippurate	500mg/dog, q12hr(PO)	250mg/cat,q12h(PO)		
Methenamine mandelate	10-20mg/kg, q8-12hr(PO)			
Methicillin				25 mg/kg, q4-6 hr(IV)
Methimazole		1.25mg-2.5mg/cat, q12hr then adjust by monitoring T4 1.25mg -2.5mg methimazole is administered twice daily(PO)		
DL-Methionine	150-300mg/kg/day (PO)	1-1.5g/cat (PO) added to food each day	50 mg/kg ,q 24 hr(PO)	20-50 mg/kg ,q24 hr(PO)
Methocarbomol	44mg/kg on first day then 22-44 mg/kg(q8hr)(PO)		110 mg/kg ,SD(IV)	5-55 mg/kg , q6 hr(IV) 40-60 mg/kg, q24 hr(PO)
Methohexital	3-6mg/kg (IV) give slowly to effect			
Methotrexate	2.5-5mg/m^2 (PO) q48h (dose depends on specific protocol) or 0.3-0.5mg/kg, Once/week(IV)	0.8mg/kg, q 2-3weeks(IV)		
Methoxamine	200-250µg/kg(IM) or 40-80 µg /kg(IV)			
Methoxyflurane	Induction 3% Maintenance 0.5-1.5%			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Methylene blue 0.1%	1.5mg/kg (IV Slowly)		4-15 mg/kg,SD (IV)	NR
Methyl prednisolone	0.22-0.44mg/kg, q12-24hr(PO)			
Methyl prednisolone acetate	1 mg/kg (or 20-40 mg/dog) Q 1 to 3 wk(IM)	10 to 20 mg /cat, q1-3 week(IM)		0.2 mg/kg,SD, as necessary (IM) 0.1 mg/kg,SD (IV)
Methyl prednisolone sodium succinate	Emergency 30 mg /kg (IV), and repeat at 15 mg /kg in 2-6 hr			0.5-1 mg/kg , q24 hr(PO) 0.5-1 mg/kg , q24 hr (IV) 10-20 mg/kg (shock) SD(IV)
Methyltestosterone	5 to 25 mg/ dog, q24-48 hr(PO)	2.5 to5 mg/ cat ,q 24-48 hr(PO)		
Metoclopramide	0.2 to 0.5 mg/ kg, q6-8h,(IV,IM,PO); IV loading dose at 0.4 mg/kg followed by 0.3 mg/kg/hr IV		0.1-1 mg/kg, q6-12h (IV,IM)	0.02-0.25 mg/kg, q6-8 hr(IV)
Metoprolol tartrate	5 to 50 mg/dog or 0.5-1 mg/kg, q8 h(PO)	2 to15 mg /cat q 8 hr(PO)		
Metronidazole and metronidazole benzoate	For anaerobes: 15 mg /kg,q12 hr(PO) Or 12 mg/kg q 8 hr(PO) For giardia 12-15 mg/kg ,q12h for 8 days	10 to 25 mg /kg, q24 hr(PO) For giardia 17mg/kg (approximately 1/3 of 250 mg tablet)q24h for 8 days		15-25 mg/kg, q8-12hr (IV, PO)
Mexiletine	5 to 8 mg/kg q 8-12 hr(usually in combination with beta blocker) (PO)	Not to be used		
Mibolerone	0.45 to 11.3 kg -30 microgram;11.8-22.7 kg -60 microgram;23-45.3 kg -120 microgram;>45.8 kg-180 microgram or approximately 2.6-5 microgram/kg /day PO	Safe dose not established		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Midazolam	0.1 to 0.25 mg /kg(IV,IM)	0.05 mg/kg (IV) or 0.3-0.6 mg/kg (combine with 3 mg/kg ketamine)(IV)		0.011-0.044 mg/kg,SD(IV)
Milbemycin oxime	Microfilaride-0.5 mg/kg;(PO) Demodex-2mg/kg q 24 h for 60-120 days(PO) Heartworm prevention:0.5 mg/kg q 30 days(PO)	2mg/kg q30 days (PO)		
Mineral oil	10 to 50 ml/dog, q 12 hr(PO)	10 to 25 ml/cat ,q 12 hr(PO)	8 ml/kg,SD (PO)	Adults: 10 ml/kg, SD (PO)
Minocycline hydrochloride	5 to 10 mg /kg,q 12 hr(PO)	8.8 mg/kg,once daily or 50 mg /cat PO once daily(PO)		3 mg/kg, q12 hr(PO)
Misoprostol	2- 5 µg/kg q6-8h (PO) Atopic dermatitis: 5 µg/kg q 8 h (PO)	Dose not established		1-4µg/kg, q8-12hr(PO)
Mitotane	For pituitary dependent hypercorticism: 50 mg/kg/day (in divided doses)PO for 5-10 days then 50 to 70 mg /kg/wk (PO) For adrenal tumor: 50 to 75 mg /kg /day for 10 days, then 75-100 mg/kg/wk (PO)			
Mitoxantrone	6 mg/m^2,q21 days (IV)	6.5 mg / m^2,q21 days (IV)		
Monensin			1 mg/kg, q24 hr (PO)	
Morantel tartarate				8-10 mg/kg,SD (PO)
Morphine	0.1-1mg/kg (dose is escalated as needed for pain relief),q4-6 h (IM,IV,SC); 0.5 mg/kg, q2h (IV,IM,CRI)	0.1 mg/kg,q 3-6 hr(IM,SC)	Sheep and goat: 1-10 mg TD,SD (IM)	0.1-0.7mg/kg,SD (IM, Slow IV)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
	CRI: 0.2 mg/kg followed by 0.1 mg/kg/h (IV) Epidural:0.1 mg/kg			
Moxalactam				50 mg/kg, q8hr (IM,IV)
Moxidectin	Heartworm prevention:3 microgram /kg Endoparasites:25-300 microgram/kg Demodex:400 microgram /kg/day upto 500 microgram /kg/day for 21-22 wks		Cattle: 0.2 mg/kg,SD (SC,PO) Cattle: 0.5 mg/kg,SD (Topical) Sheep: 0.2 mg/kg,SD (PO) Goat: 0.2-0.5 mg/kg,SD (SC,PO)	0.4 mg/kg,SD (PO)
Moxifloxacin	10 mg/kg,q24 hr(PO)			
Mycophenolate	10-20 mg/kg, q8hr (PO)	10 mg/kg,q12 hr (PO)		
Nafcillin				10 mg/kg, q6 hr (IM)
Naloxone	0.01-0.04 mg/kg,as needed to reverse opiate (IV,IM,SC)			0.01-0.05 mg/kg,SD (IV)
Naltrexone	For behaviour problems:2.2 mg/kg,q12 hr (PO)			
Nandrolone decanoate	1-1.5 mg/kg/week (IM)	1 mg/cat/week (IM)		
Naproxen	5 mg initially,then 2 mg/kg ,q 48 hr	Not recommended		5-10 mg/kg, q12-24 hr (PO)
Neomycin	10 -20 mg/kg,q 6-12 hr (PO)		3-6 mg/kg, q6-12 hr (PO) 88 mg/kg, q8 hr (SC)	2- 6 mg/kg, q6-12 hr (PO) 4.4 mg/kg, q8-12 hr (IV)
Neostigmine bromide and neostigmine methylsulfate	2 mg/kg /day (in divided doses, to effect) (PO) Injection:Antimyasthenic:10µg/kg (IM,SC) Antidote for non depolarizing neuromuscular block:40µg/kg (IM,SC)		0.02 mg/kg, SD (SC)	0.004-0.02 mg/kg, SD (SC)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
	Diagnostic aid for myasthenia gravis:40µg/kg (IM) or 20µg/kg (IV)			
Netilmicin				2 mg/kg, q8-12 hr(IM,IV)
Netobimin			7.5 mg/kg,SD(PO)	
Niclosamide				100 mg/kg,SD(PO)
Nizatidine				6.6 mg/kg, q8h (PO)
Nitazoxanide				25 mg/kg for days 1-5 days, then 50 mg/kg for days 6-28, q24hr (PO)
Nitenpyram	1 mg/kg daily as needed to kill fleas(PO)			
Nitrofurantoin	10 mg/kg/day divided into four daily treatments,then 1 mg/kg at night(PO)			2.5-5 mg/kg,q8h (PO)
Nitroglycerin ointment	4-12 mg(upto 15 mg),q 12 hr (TOPICAL)	2-4 mg,q 12 hr (Or ¼ inch of ointment per cat) (TOPICALLY)		15 mg TD, q24h (topical over each digital artery)
Nitroprusside	1-5,upto a maximum of 10 µg/kg/min(IV infusion) Start with 2µg/kg/min and increase gradually until desired blood pressure is attained. At doses more than 10µg/kg /min seizures are more likely			
Nitroxynil			10-15 mg/kg,SD(SC)	
Nizatidine	5 mg/kg,q24 hr(PO)			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Norepinephrine				0.1-1.5µg/kg/min, S D(IM)
Novobiocin			400 mg TD(dry cow),SD(IM) 150 mg TD(lactating cow),Q 24 hr(IM)	
Nystatin				250000-1000000,SD(I U)
Omeprazole	20 mg/dog,daily q24h (PO); 0.7mg/kg q24h	0.5 to 0.7 mg/kg,q 24 h(PO)		1-4 mg/kg,q24 hr(PO)
Ondansetron	0.5 to 1 mg/kg30 minutes before administrat ion of cancer drugs(IV,PO)			
Orbifloxacin	2.5-7.5mg/kg, (PO) Once daily	7.5mg/kg (PO) once daily		
Oxacillin	22 to 40 mg/kg, q8h(PO)			25-50 mg/kg, q8-12 hr (IM,IV)
Oxazepam		2.5mg/cat(PO)		
Oxfendazole			4.5 mg/kg,SD(PO)	10 mg/kg,SD(PO)
Oxibendazole			10-20 mg/kg,SD (PO)	10-15mg/kg,SD,(PO)
Oxtriphylline	47 mg/kg, q12h,(PO)			
Oxybutynin chloride	5 mg/dog, q6- 8 h (PO)			
Oxyclozanide			10-15 mg/kg,SD (PO)	
Oxymetholone	1-5mg/kg/day (PO)			
Oxymorphone	Analgesic: 0 .1-0.2mg/kg (IV, SC,IM) Redose with 0.05-0.1 mg/kg(q1-2h)	0.03mg/kg bolus followed by 0.02mg/kg q2h Preanaesthetic:0.025-0.05 mg/kg (IM,SC)		0.01-0.02 mg/kg,SD,(IM,I V)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Oxytetracycline	7.5-10mg/kg (q12h),(IV) 20mg/kg,q 12h,(PO)		5-20 mg/kg, q12-24h (IV, IM)	6.6-10 mg/kg, q24h, (Slow IV) Foals: 10-20 mg/kg, q24h(Slow IV)
Oxytocin	5-20 U/dog,(SC,IM) Repeat every 30 min for primary inertia	2.5-3U/cat(SC,IM)	0.05-0.1 mg/kg (retained placenta), q1-1.5h (IM) 0.025-0.05 mg/kg (milk let down),SD(IV)	Induction of foaling: 0.05-0.1 mg/kg,SD (IM) Retained placenta: 0.01-0.02mg/kg, q1-1.5h (IM)
Pamidronate	2 mg/kg (IV,SC) Cholecalcif erol toxicosis:1.3-2mg/kg for 2 treatments			
Pancrelipase	Mix 2 TSP powder with food per 20 kg body weight or 1-3 TSP/0.45 kg of food 20 min before feeding			
Pancuronium bromide	0.1mg/kg(IV)		Cattle: 0.04 mg/kg then 0.008mg/kg,SD (IV) Sheep: 0.025 mg/kg then 0.005 mg/kg,SD (IV)	0.04-0.066 mg/kg,SD(IV)
Pantoprazole	0.5mg/kg, q24 (IV) or 0.5-1mg/kg IV infusion	0.5mg/kg,q24h,(IV),		1.5 mg/kg,q24 hr,(IV,PO)
Paregoric	0.05-0.06mg/kg, q12h,(PO)			
Paromomycin				100 mg/kg,q24 hr (PO)
D-Penicillamine	10-15mg/kg, q12h,(PO)		52 mg/kg,q24 hr,(PO)	3-4 mg/kg,q6 hr (PO)
Penicillin G benzathine	24000 U/kg, q48h,(IM)		44,000-66000 IU/kg,48-72 hr,(IM,SC)	10000-40000 IU/kg,q48-72 hr,(IM)
Penicillin G potassium; penicillin G sodium	20000-40000 U/kg, q6-8h, (IV,IM)			10 000-50 000 IU/kg ,6-8 hr, (IV,IM)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Penicillin G procaine	20000-40000 U/kg,(q12-24h),(IM)		10 000-60 000 IU/kg,12-24 hr,(IM,SC)	20 000-50 000 IU/kg,12-24 hr,(IM)
Penicillin V	10 mg/kg, q8h,(PO)			
Penicillin V, potassium				66 000-110 000 IU/kg ,6-8 hr,(PO)
Pentobarbital	25-30mg/kg (IV) to effect or 2-15mg/kg IV to effect followed by 0.2-1mg/kg/hr (IV)		30 mg/kg (to effect),SD,(IV) 120-200 mg/kg for euthanasia,SD,(IV)	2-20 mg/kg(to effect),SD,(IV) 120-200 mg /kg for euthanasia,SD ,(IV)
Pentazocine				0.33 mg/kg,SD (IV,IM,SC)
Pentosan sulfate				250 (TD),7days,(IA) 3 mg/kg,7days (IM)
Pergolide				0.002-0.004 mg/kg,q24 h (PO)
Perphenazine				0.3-0.5 mg/kg,q12 h (PO)
Pentoxifylline	For use in canine dermatology and for vassculitis 10mg/kg q12h (PO) upto 15mg/kg, q8h	¼ of 400 mg/ kg, q8-12 h,(PO)		8 mg/kg,q12 hr,(PO)
Phenobarbital	2-8mg/kg, Q12,(PO)	2-4 mg/kg, Q12h,(PO)	10 mg/kg, 24 hr(PO)	5-25 mg/kg,SD (IV) 1-5 mg/kg,q12hr,(PO)
Phenothiazine				55 mg/kg,SD (PO)
Phenoxybenzamine	0.25mg/kg (PO) q8-12h or 0.5mg /kg, q24 (PO)	2.5mg/cat, q8-12h (PO) or 0.5mg/cat (PO) q12h		0.6 mg/kg,q6-8hr,(IV) 0.6-1.2 mg/kg,q12 hr,(PO)
Phentolamine	0.02-0.1 mg/kg,(IV)			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Phenylbutazone	15-22mg/kg (PO) IV q8-12h (44 mg/kg/day) (800mg/dog maximum)	6-8mg/kg IV, PO, q12h	4 mg/kg, q24 hr,(IV) 10-20 mg/kg ((loading dose)),q24-48 hr then 5-10 mg/kg, q24-48h (PO)	2-4.4 mg/kg,q12-24hr,(PO,IV)
Phenylephrine	0.01 mg/kg, q15min (IV)	0.1 mg/kg, q15min,(IM,SC)		
Phenylephrine hydrochloride				0.02 -0.04 mg/kg,SD, IV (over 10 min)
Phenylpropanalamine	1mg/kg, q12h; 1.5-2mg /kg, q8h (PO)			
Phenytoin	Antiepileptic:20-35 mg/kg, Q8h, PO Antiarrhythmic: 30mg/kg, Q8h(PO) 10 mg/kg(IV) over 5 min			
Phenytoin sodium				Seizures: 5-10 mg/kg then 1-5 mg/kg SD, (IV)q4-8 hr(IV,IM,PO) Rhabdomyolysis: 10-12 mg/kg,q12 hr,(PO) Arrhythmias: 10-22 mg/kg,q12 hr, (Slow IV, IM)
Physostigmine	0.02 mg/kg, q12(IV)			0 .1-0.6 mg/kg,SD(IM, IV)
Phytonadione (Vitamin K1)			0.5-2.5 mg/kg,SD,8hr(IV(slow),IM)	0.5-2.5 mg/kg,SD (Slow IV)
Pimobendan	0.05mg/kg/day in divided treatments, q12h (PO)	0.25mg/kg, q12h (PO)		
Piperacillin	40 mg/kg , Q6h(IV,IM)			15-50 mg/kg,q6-12 hr(IV,IM)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Piperazine	44-66mg/kg(PO)			110-200 mg/kg,SD(PO)
Pivampicillin sodium				20 mg/kg,q12 hr(PO)
Pirbuterol				0.001-0.002 mg/kg,q12-24 hr (IH)
Pirlimycin			50mg/kg TD,q12-24hr (IMM)	
Piroxicam	0.3mg/kg, q48h (PO)	0.3mg/kg, Q24h(PO)		
Polyethylene glycol electrolyte solution	25 ml/kg , Q2-4hr(PO)			
Poloxalene			110 mg/kg, q24 hr(PO)	
Polymyxin B			6000 mg/kg,SD(IV)	6000 mg/kg,SD(IV)
Plicamycin	Antineoplastic:25-30µg/kg/day for 8-10 days (IV); Antihypercalcemic: 25 microgram/kg/day, Over 4hr(IV)	Not recommended		
Polysulfated glycosaminoglycan (PSGAG)	4.4mg/kg , Twice weekly up to 4 weeks (IM)			0.5mg/kg, q96 hr(IM) 0.25 mg/kg, q96hr(IA)
Potassium bromide (KBr)	30 – 40 mg/kg , Q 24h(PO)	30-40mg/kg,Q24h(PO)		20-40 mg/kg,q24 hr(PO)
Potassium chloride (KCl)	0.5 mEq potassium /kg/day or supplement 10-40mEq /500ml of fluids depending on serum potassium. (PO)			
Potassium citrate	0.5mEq / kg / day (PO)			
Potassium gluconate	0.5mEq /kg , Q 12-24h(PO)	2.8mEq/day , Twice daily(PO)		
Potassium iodide		30-100mg/cat , Daily (single or divided dose) for 10-14days	1.5 mg/kg, q24 hr(IV)	4-40 mg/kg,q24 hr(PO)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Pralidoxime chloride (2 -PAM)	20mg/kg ,Q 8-12 h(IV slow or IM)		25-50 mg /kg,SD(IV slow)	20-50 mg/kg, q4-6 hr(IV)
Praziquantel	<6.8 kg, 7.5mg/kg,once; >6.8kg, 5mg/kg,once(PO) <2.3kg, 7.5mg/kg,once; 2.7-4.5kg, 6.3mg.kg,once >5kg, 5mg/kg once (IM,SC)	<1.8kg, 6.3mg/kg once >1.8 kg ,5mg/kg(PO) once 5mg/kg (IM,SC)	10-15 mg/kg, SD(PO)	1-2 mg/kg,SD(PO)
Prazosin	0.5 and 2mg / dog, q 8-12h (PO)	1mg/15kg, q8-12 h(PO)		
Prednisolone	Anti-inflammatory = 0.5 -1mg/kg, q12-24h(IV,IM,PO) initially then taper to q48h Immunosuppressive: 2.2-4.4mg/kg/day (IV,IM,PO) initially then taper to 2-4mg/kg q48h Replacement therapy: 0.2-0.3mg/kg/day (PO)	Cat often requires 2 times the dog dose	1-4 mg/kg,SD,q24(IV)	0.2-4.4 mg/kg, q12-24 hr(IM,PO)
Prednisolone sodium succinate	Shock: 15-30mg/kg (IV) repeat in 4-6h CNS trauma: 15-30mg/kg (IV), taper to 1-2mg/kg, q12h			Adult: 50-100 mg/kg,SD(IV)
Primidone	8-10mg/kg, Q8-12h then adjusted via monitoring to 10-15mg/kg, Q8 (PO)			10-20 mg/kg,q6-12 hr(PO)
Primor (ormetoprim +sulphadimethoxine)	27mg/kg on 1st day, followed by 13.5mg/kg, q24h (PO)			
Procainamide	10-30mg/kg, q6h (max. 40mg/kg) (PO) 8-20mg/kg (IV,IM) 25-50µg/kg/min IV infusion	3-8mg/kg , Q6-8h(PO,IM)		0.5 mg/kg to total dose of 4 mg/kg,q10 min(IV)
Prochlorperazine	0.1 to 0.5mg/kg, q6-8h (IM, SC)			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Promazine				0.25-1 mg/kg,SD(IV) 1-2 mg/kg,SD(PO)
Promethazine	0.2 to 0.4mg/kg upto maximum dose of 1mg/kg, q6-8h (IV,IM,PO)			
Propafenone				0.5-1 mg/kg,SD(IV)
Propantheline bromide	0.25 to 0.5mg/kg, q8-12h (PO)			0.014 mg/kg,SD(IV)
Propiomazine	1.1 to 4.4mg/kg (PO); 0.1 to 1.1 mg/kg(q12-24h) IM,IV (range of dose depends on degree of sedation needed)			
Propofol	6.6mg/kg IV slowly over 60 seconds; constatnt rate IV infusions have been used at 5 mg/kg slowly IV, followed by 100-400 µg/kg/min IV			Induction: 2.4 mg/kg Maintenance: 0.3 mg/kg, IV infusion
Propranolol	20 to 60 µg/kg over 5 to 10 mins (IV); 0.2 to 1 mg /kg,q8hrs (PO) (titrate dose to effect)	0.4 to 1.2mg/kg (2.5 to5mg/cat) q8hrs (PO)		0.03-0.15 mg/kg, q8hr (IV) 0.4-0.8 mg/kg,q 8 hr(PO)
Propylthiouracil	11mg/kg,q12hr (PO)			
Propylene glycol			Cattle: 110-225 ml TD, q24 hr(PO) Sheep: 110 ml TD, q24 hr(PO)	
Protamine sulphate			0.2 mg/kg, SD(IV)	
PGF2alpha	Pyometra: 0.1 to 0.2mg/kg, once daily for 5 days (SC) Abortion: 0.025 to 0.05 mg (25-50 µg)/kg, q12 hr (IM)	Pyometra: 0.1 to 0.25mg/kg, once daily for 5 days (SC) Abortion: 0.5 to 1mg/kg for two injection (IM)		0.02 mg/kg,SD(IM)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Pseudoephedrine	0.2 to 0.4 mg/kg (or 15-60 mg/dog), q8-12h (PO)			
Psyllium	1tsp /5-10 kg (added to each meal)			500 mg/kg, q12-24 hr(PO)
Pyrantel pamoate	5mg /kg (PO) once and repeat in 7-10 days	20mg/kg, Once(PO)	25 mg/kg,SD(PO)	6.6 mg/kg,SD(PO)
Pyrantel tartarate				2.6 mg/kg,SD(PO)
Pyridostigmine bromide	Antimyasthenic: 0.02-0.04 mg/kg (IV) q2h or 0.5 to 3 mg/kg, q8-12h (PO) or CRI of 0.01-0.03 mg/kg/hr Antidote (non depolarizing muscle relaxant): 0.15-0.3 mg/kg (IM, IV)	0.1 to 0.25mg/kg, q24h (PO)		
Pyrilamine maleate			0.55mg/kg,SD (IV,IM)	0.8-1.3 mg/kg, q6-12 hr (Slow IV,IM,SC)
Pyrimethamine	1mg/kg,q24h for 14 to 21 days (PO) (5 days for *Neospora caninum*)	0.5 to 1mg/kg, q24h for 14 to 28 days (PO)		1-2 mg/kg, q24 hr(PO)
Quinidine gluconate	6 to 20 mg /kg,q6h (IM); 6 to 20 mg/kg, q6-8h (of base) (PO)		50 mg/kg (over 6 hr) (IV) 210 mg/kg loading dose then 180 mg/kg, q6 hr (PO)	22 mg/kg, q2-4 hr(PO) 0.5 -2.2 mg/kg, q10 min until conversion to sinus rhythm or suppression of arrhythmia(IV)
Quinidine sulfate	6-20 mg/kg, q6-8h (of base) (PO); 5-10 mg /kg (IV)			
Quinidine polygalacturonate	6-20 mg/kg, q6h (of base) (PO)			
Racemethionine	150-300mg/kg/day(PO)	1 to 1.5 gm/cat (PO) added to food each day		
Ranitidine	2 mg/kg, q8h(IV PO)	2.5 mg/kg (IV)	Calves: 50 mg/kg ,q8	6.6 mg /kg, q6-12 hrs(PO)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
		q12h, 3.5 mg/kg (PO) q12h	hr(PO)	1.5 mg/kg, q6-12 hr(IV)
Reserpine				0.002-0.008 mg/kg,q24 hr (PO)
Retinoids	See Isotretinoin (Accutane), retinol (aquasol-A), or etretinate (Tegison).			
Retinol	See vitamin A (Aquasol-A)			
Riboflavin	See vitamin B2			
Rifampin	5 mg/kg, q12h (PO)			5-10 mg/kg ,q12 hr(PO)
Ringer's solution	55-65 ml/kg/day IV SC or IP; 50ml/kg/hr IV for severe dehydration			
Romifidine				0.04-0.12mg/kg, SD(Slow IV,IM)
Ronidazole	Dose not established	30 mg/ kg/day for 2 weeks (PO). Dose of 30mg/kg per day may be divided into twice daily treatments		
Salicylate	See Aspirin, acetylsalicylic acid			
Salmeterol				0.0005-0.001 mg/kg,q6-12 hr (IH)
Selegiline (deprenyl)	Begin with 1 mg/kg, q24h(PO) if there is no response within 2 months, increase dose to max. of 2 mg/kg PO q24h	0.25-0.5 mg/ kg,q12-24 h (PO)		
Senna	Syrup: 5-10ml/dog q24h; Granules ½ to 1 tsp/dog(q24h) PO with food	Syrup: 5 ml/cat q24h; Granules ½ teaspoon/cat,q24 h with food		
Septra(sulfamethoxazole+ trimethoprim)	See Trimethoprim/Sulfonamides			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Sildanefil	0.5- 1mg/ kg, q12h (PO); higher dose of 2-3 mg/kg q8h may be needed in some cases	1 mg/kg q8h PO		
Silymarin	30 mg/ kg/day(PO)			
Sodium bicarbonate	Acidosis 0.5-1 mEq/kg (IV) Renal failure: 10 mg/ kg, q8-12h (PO) Alkalization of urine: 50 mg/kg (1 tsp is approximately 2 g), q8-12h (PO)			
Sodium bromide	Same as potassium bromide, except dose is 15% lower (30 mg/kg potassium bromide is Equivalent to 25 mg/ kg sodium bromide)			
Sodium chloride 0.9%	15-30 ml/kg/hr(IV)			
Sodium chloride (7%)			4ml/kg,SD(IV)	4ml/kg,SD(IV)
Sodium chloride 7.5%	2-8 ml/kg(IV)			
Sodium sulphate			1-2 mg/kg,SD (PO)	1-2 mg/kg,SD,(PO)
Somatropin	See growth hormone			
Sodium thiomalate	See gold sodium thiomalate			
Sodium thiosulphate			Cyanide poisoning: 660 mg/kg, SD(IV) Copper poisoning: 1mg/kg, q24 hr(PO)	30-40mg/kg,SD(IV)
Sotalol	1-2 mg/kg, q12h (PO) (one can start with 40 mg/dog	1-2 mg/kg(PO) q12h		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
	q12h, then increase to 80 mg if no response)			
Spectinomycin			10-15 mg/kg, q24hr(SC)	20 mg/kg, q8 hr(IM)
Spironolactone	2-4 mg/kg /day(or 1-2 mg/kg q12h) (PO)	Questionable efficacy, may produce skin lesions		
Stanozolol	1-4 mg/ dog, q12h (PO); 25-50mg/dog/wk (IM)	1 mg/cat,q12h(PO); 25 mg/cat/wk (IM)	2 mg/kg,SD(IM)	0.55 mg/kg,q168hr (IM)
Stilbestrol				0.3 mg/kg,SD(IM)
Streptomycin			11mg/kg,q12 hr(IM,SC)	11 mg/kg,q12 hr(IM,SC)
Succimer	10 mg/kg, q8 hr (PO) for 5 days, then 10 mg/kg (PO) q12h for 2 more weeks	10 mg/Kg, q8 hr for 2 weeks (PO)		
Succinylcholine chloride				0.09-0.11 mg/kg,SD(IV)
Sucralfate	0.5-1gm/ dog, q8-12h (PO)	0.25gm/Cat, q8-12h(PO)		10-20 mg/kg,Q 6-12 hrs(PO)
Sufentanil Citrate	2µg/kg (IV) to a max. dose of 5 µg/kg			
Sulfachlorpyridazine			88-110 mg/kg, q12-24 hr(IV) Calves: 30-50 mg/kg, q8 hrs(PO)	
Sulfadiazine	100mg/kg, q12 h(IV PO) loading dose followed by 50 mg/kg IV PO q12h			
Sulfadimethoxine	55 mg/kg PO (loading dose) followed by 27.5mg/kg PO q12h		55-110 mg/kg ,q24 hr(PO) 55 mg/kg then 28 mg/kg,q24 hr(IV)	55 mg/kg,Q 24 hr(IV)
Sulphonamide/ trimethoprim			15-30 mg/kg, q12-24 hrs(IM,IV) Pre-ruminant calves: 15-30 mg/kg, q12-24 hrs(PO)	15-30 mg/kg, q12-24hr (IV, IM, PO)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Sulfadimidine			100-200 mg/kg loading dose (SD) then 50-100 mg/kg, q24hr (IV)	
Sulphadoxine/tri methoprim			15 mg/kg,q12-24 hrs(IM,IV,SC)	15 mg/kg, q12-24 hr (IM, Slow IV)
Sulfamethoxazole	100mg/ kg PO (loading dose), followed by 50mg/kg PO q12h			
Sulfamethoxypyri dazine			20 mg/kg, q24 hrs(SC,IM,IV,IP)	
Sulfasalazine (Sulfapyridine+ Mesalamine)	10-30 mg/kg, q8-12 hr (PO)	20 mg/kg, q12hr(PO)		
Sulfisoxazole	50 mg/Kg, q8 h(PO) (urinary tract infections)			
Suxamethonium chloride			0.02 mg/kg,SD(IV)	0.1 mg/kg,SD(IV)
Taurine	500mg, q12 hr (PO)	250 mg/Cat, q12h (PO)		
Terbinafine	*Malassezia dermatitis*: 30 mg/kg/day (PO)	Dermatophytosis: 30-40mg/kg, q24h (PO)		
Terbutaline	1.25-5 mg/ dog, q8h (PO)	0.1-0.2mg/kg, q12h (PO); For acute treatment in cats: 5-10 µg/kg q4h (SC IM)		0.02-0.06 mg/kg, q6-12 hr(PO) 0.002 mg/kg,SD(IV)
Testosterone cypionate ester	1-2 mg/kg IM q2-4 wk			
Testosterone propionate ester	0.5-1 mg/ kg, 2-3 times/wk(IM)			
Tetanus antitoxin				3 IU (tetanus prophylaxix),SD(IM,IV,SC) 100 IU (treatment of tetanus),q72-120 hr(IM,IV,SC)
Tetracycline	15-20 mg/kg PO q8h; 4.4-11 mg/kg (IV IM) q8h			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Thenium Closylate	4.5 kg: 500m g noce, repeat in 2-3 wk (PO) 2.5-4.5 kg: 250 mg q12h for 1 day, repeat in 2-3 wk			
Theophylline	9mg/kg, q6-8h (PO)	4mg/kg, q8-12h (PO)		
Theophylline extended release	10mg/kg q12h (PO) of extended release tablet or capsule	20mg/kg q24-48h (PO) of extended release tablet; 25mg/kg q24-48h (PO) extended release capsule		8-12 mg/kg, q8-12 hr(PO)
Thiamine (Vitamin B1)	10-100mg/dog/day (PO) or 12.5-50mg/dog/day (SC,IM)	5-30mg/cat/day (PO) upto a max. dose of 50mg/cat/day or 12.5-25 mg/cat/day (SC,IM)	5-50 mg/kg, 12 Hr(IM,IV)	0.5-5 mg/kg, SD(IV,IM,PO)
Thiamylal sodium			4.4-8.8 mg/kg,SD(IV)	2-4 mg/kg,SD(IV to effect)
Thiabendazole			50-100 mg/kg,SD(PO)	44 mg/kg,SD(PO)
Thioguanine	40mg/m^2 q24h (PO)	25mg/m^2 q24h (PO) for 1-5 days, then repeat every 30 days		
Thiopental sodium	10-25mg/kg (IV to effect)	5-10 mg/kg (IV to effect)	8-16 mg/kg,SD(IV)	6-12 mg/kg,SD(IV)
Thiophanate			Cattle: 2.4-4.8 g TD,SD(PO) Sheep: 240-480 mg TD,SD(PO)	
Thiotepa	0.2-0.5mg/ m^2 weekly or daily for 5-10 days (IM, intracavitary or intratumor)			
Thyrotropin TSH	Collect baseline sample followed by 0.1U/kg; maximum dose is 5U (IV); collect post TSH sample at 6 hr	Collect baseline sample followed by 2.5U/cat and collect a post TSH sample at 8-12 hr (IM)		

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Thyroxine L				0.01 mg/kg,q24 hr(PO)
Ticarcillin	33-50mg/kg q4-6h (IV,IM)			50 mg/kg,q6-8 hr(IV,IM)
Tiletamine + zolazepam	6.6-10mg/kg short term (IM) or 10-13mg/kg longer procedure (IM)	10-12mg/kg minor procedure (IM) or 14-16mg/kg for Surgery(IM)		1.6-2.2 mg/kg ,SD(IV)
Tilmicosin			10 mg/kg,q72 hr(SC)	
Tobramycin	9-14mg/kg q24h(IM,IV,SC)	5-8mg/kg q24h(IM,IV,SC)		4 mg/kg, every 24 hrs(IV, IM)
Tocainide	15-20mg/kg q8h (PO)	No dose established		
Toluene	267 mg/kg PO (of toluene), repeat in 2-4 wk			
Tocopherol acetate(vitamin E)				10-15 IU/kg, q24 hr(PO)
Tolazoline				4 mg/kg ,SD(IV slow)
Tramadol hydrochloride	5mg/kg q6-8h (PO)	Start with 2mg/kg q12h and increase gradually to 4mg/kg q8-12h (PO). Cats are more prone to adverse reaction		
Trandolapril	Not established for dogs; human dose is 1mg/person/day to start, then increase to 2-4mg/day			
Trenbolone acetate			140-200 TD,SD(SC)	
Triamcinolone acetonide	0.1-0.2mg /kg repeat in 7-10 days (IM,SC) Intralesional: 1.2-1.8mg or 1 mg for every cm diameter of tumor q2 weeks		0.02-0.04 mg/kg,SD(IM)	0.1-0.2 mg/kg,SD(IM, SC) 6-18 TD,SD(IA)

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Triamcinolone	Anti-inflammatory: 0.5-1mg/kg q12-24h, then taper dose to 0.5-1mg/kg q48h (PO)			
Triamterene	1-2mg/kg q12h (PO)			
Trichlorphon				35-40 mg/kg,SD(PO)
Triclabendazole			12 mg/kg,q8-10 weeks(PO)	
Trientine hydrochloride	10-15mg/kg , q12h (PO)			
Trifluoperazine	0.03 mg/kg, q12h (IM)			
Triflupromazine	0.1-0.3 mg/kg IM, q8-12h			
Trilostane	3-6mg/kg, q24h (PO) adjust dose with cortisol measurements	6mg/kg, q24h (PO) and gradually increase as needed to 10mg/kg, q24h (PO)		0.4-1 mg/kg ,q24 hr(PO)
Trimeprazine tartrate	0.5mg/kg, q12h (PO)			
Trimethobenzamide	3mg/kg, q8h (IM,PO)	Not recommended		
Trimethoprim + sulfonamides	15mg/kg, q12h (PO) 30mg/kg , q12-24h (PO) For *Toxoplasma*: 30 mg/kg PO q12h			
Tripelennamincitrate and Tripelannamine hydrochloride	1mg/kg, q12 h or 0.25ml/5kg (PO)		1.1 mg/kg, q6-12 hrs (IV,IM)	1.1 mg/kg,q6-12 hr(IM)
Tubocurarine hydrochloride			Cattle: 0.06 then 0.01 mg/kg SD to effect(IV) Sheep: 0.04 then 0.01 mg/kg, SD to effect(IV)	0.3 then 0.05 mg/kg ,SD(IV)
Tulathromycin			2.5 mg/kg ,SD(SC)	

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Tylosin	7-15mg/kg, q12-24 h (PO) Colitis: 10-20mg/kg, q8h with food initially, then increase interval to q12-24h (PO)	7-15mg/kg, q12-24 h (PO)	18 mg/kg, q24 hr(IM)	Not recommended
Urofollitropin	75U/day for 7 days(IM)			
Ursodiol	10-15mg/kg q24 h (PO)			
Valproic acid, divalproex	60-200mg/kg q8h or 25-105mg/kg/day (PO) when administered with phenobarbital			
Vancomycin	15mg/kg q6-8h (IV infusion)	12-15mg/kg q8 h (IV infusion)		7.5mg/kg, q8hr, IV
Vasopressin	10U/animal (IV,IM)			
Verapamil	0.05mg/kg q10-30 min (IV) (maximum cumulative dose is 0.15 mg/kg)			0.025-0.5mg/kg SD to effect IV
Vinblastine	2mg/m² once /week(IV)			
Vincristine	Antitumor: 0.5-0.7mg/m² (0.025-0.05 mg/kg) once/wk For thrombocytopenia: 0.02mg/kg once/week(IV)			
Vitamin A	625- 800U/kg q24h (PO)			
Vitamin B2 (Riboflavin)	10 to 20 mg /day(PO)	5 to 10 mg /day(PO)		
Vitamin B_{12} (cyanocobalamin)	100 to 200 µg/day(PO)	50 to 100 µg/day(PO)		
Vitamin C (ascorbic acid)	100 to 500 mg /day			

SPECIES	DOG	CAT	RUMINANTS (Cattle, Sheep, Goat)	HORSE
Name of the Drug	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal	Dose & Dosage Internal
Vitamin E (Alpha tocopherol)	100 to 400 U, q12h (PO) (400-600 U PO q12h for immune mediated skin disease)			6-10 IU/kg, q24 hr(PO)
Vitamin K1 (phytomenadione)	Short acting rodenticide: 1mg/kg/day for 10 to 14 days (IM,SC) Long Acting rodenticide: 2.5 to 5 mg/kg/day and upto 6 wk for 3to 4 wk (IM,SC,PO)			
Warfarin	0.1 to 0.2 mg/kg , q24h (PO)	Thromboembolism: Start with 0.5 mg /cat/day and adjust dose based on clotting time assessment		0.02 mg/kg then slowly increasing to effect, q24 hr(PO)
Xylazine	1.1 mg/kg(IV) 2.2 mg /kg(IM)	5 to 10 mg/kg q12h (PO,SC)	0.01-0.05 mg/kg,SD (IV) 0.02-0.10 mg/kg,SD (IM)	1.1 mg/kg,SD(IV) 2.2 mg/kg,SD(IM)
Yohimbine	0.11 mg/kg (IV) or 0.25-0.5 mg/kg (SC, IM)		0.125 mg/kg ,SD(IV)	0.05-0.2 mg/kg,SD(Slow IV, IM)
Zeranol			36-72mg TD ,SD(SC)	
Zidovudine		5 to 10 mg/kg, q12h (PO,SC)		
Zonisamide	3 mg/kg, q8h (PO); it has also been administered to dogs at 10 mg/kg q12h PO	Dose not established		

CHAPTER 6

DRUG CONTRAINDICATIONS
DRUGS NOT RECOMMENDED FOR USE IN DOGS & CATS

DOGS	CATS
1. Ivermectin (Collie breeds of dog) 2. Acetaminophen 3. Ibuprofen 4. Naproxen	1. Acetaminophen (Paracetamol) 2. Apomorphine 3. Azathioprine 4. Benzocaine 5. Cisplatin 6. Propylthiouracil 7. Phenytoin 8. Scopolamine 9. Sodium phosphate enemas 10. Permethrins (high concentration products)

DRUGS NOT RECOMMENDED FOR USE IN YOUNG ANIMALS

1. Ciprofloxacin
2. Praziquantel
3. Permethrins
4. Ivermectin
5. Dicyclomine
6. Lactulose
7. Tetracyclines
8. Aminoglycosides
9. Chloramphenicol
10. Metronidazole
11. Sulpha
12. Trimethoprim

DRUGS TO BE AVOIDED/ USED WITH CAUTION IN LIVER AND RENAL DISEASES

Drug class	Avoid /hepatotoxic or use with caution in liver disease	Avoid / nephrotoxic or use with caution in renal disease
Antimicrobial drugs	Chloramphenicol Chlortetracycline Erythromycin estolate Flucytosine Griseofulvin Ketoconazole Lincosamides Macrolides Metronidazole Sulphonamide Trimethoprim Sulphonamides Tetracyclines	Aminoglycosides Amphotericin Fluoroquinolones Lincomycin Naficillin Nalidixic acid Nitrofurantoin Polymyxins Sulphonamide Trimethoprim Sulphonamides Tetracyclines (except doxycycline)
Anesthetics (general/ local/sedatives/anticonvulsants)	Anticonvulsants Barbiturates Chlorpromazine Diazepam Halogenated anaesthetics Ketamine Lidocaine Propofol	Acepromazine Chlorpromazine Ketamine Methoxyflurane Procainamide
Cardiac drugs	ß blockers Lidocaine Quinidine	Angiotensin converting enzyme inhibitor Cardiac glycosides Procainamide

Drug class	Avoid /hepatotoxic or use with caution in liver disease	Avoid / nephrotoxic or use with caution in renal disease
Diuretics	-	Spironolactones Thiazides
Anti-inflammatories/analgesics	Butorphanol Corticosteroids Meclofenamic acid Phenylbutazone Polysulfated glycosaminoglycan	Non steroidal anti-inflammatories Pethidine Polysulfated glycosaminoglycan
Cytotoxic drugs	Doxorubicin	Cisplatin Doxorubicin Fluorouracil Methotrexate
Miscellaneous	Doxapram Heparin Suxamethonium	Allopurinol Doxapram Gallamine Piperazine

DRUGS CONTRAINDICATED IN PREGNANCY

Drug class	Contraindicated in pregnancy
Antimicrobials	Chloramphenicol Nitrofuranoin Streptomycin Gentamicin Amikacin Tetracycline Trimethoprim Sulphonamides

Drug class	Contraindicated in pregnancy
	Amphotericin B Griseofulvin Flucytosine Erythromycin Metronidazole
Anti-inflammatory drugs	NSAID (aspirin, phenylbutazone, indomethacin) Glucocorticoids Dimethyl sulfoxide
Anticonvulsants	Phenobarbital Primidone Phenytoin Diazepam
Antineoplastic drugs	All Cytotoxic drugs Propylthiouracil Diazoxide
Miscellaneous	Androgens Acetazolamide Anticoagulants Chlorpromazine Cholinesterase inhibitors Cyclizine EDTA Estrogens Gold salts Iodides Isoproterenol Lithium salts Mepivacaine

Drug class	Contraindicated in pregnancy
	Meclizine Methoxamine Meperidine Prochlorperazine Propranolol Phenylephrine Quinine Thiazides Tolbutamide Vitamin A (Large dose) Vitamin D (Large dose) Vitamin K

CHAPTER 7

COMMON POISONING AND ANTIDOTES

LIVESTOCK

S. No	Poisoning	Antidotes
1.	**Organophosphorus compounds and carbamates**	• Atropine - 0.25 mg/kg BW in cattle and 1 mg/kg BW in sheep. Repeated at 4-5-hourly intervals, continued over a period of 24-48 hours • 2-pyridine aldoxime methiodide-50-100 mg/kg BW given intravenously for OPC poisoning
2.	**Urea**	• Orally vinegar (0.5-1 L to a sheep, 4 L to a cow), or 5% acetic acid • Cold water (10-30 L for adult cattle)
3.	**Cyanogenic glycoside Poisoning**	• Standard primary treatment - injection of a mixture of 5 g sodium nitrite, 15 g sodium thiosulfate in 200 mL water for cattle; 1 g sodium nitrite, 3 g sodium thiosulfate in 50 mL water for sheep, and field experience with it has been very good • Sodium thiosulfate in heavier dose 660 mg/kg BW • Animals exposed but no signs - 30 g of sodium thiosulfate orally to cattle and are repeated at hourly intervals
4.	**Nitrate and nitrite Poisoning**	• Methylene blue - 1-2 mg/kg BW, injected as a 1 % solution

PET ANIMALS

S. No.	Poison	Antidotes
1.	Acetaminophen	N-acetylcysteine
2.	Alkaloids	Tannic acid
3.	Amitraz	Atipamazole, Yohimbine
4.	Anticoagulant rodenticide	Vitamin K1
5.	Arecoline	Atropine, tannin
6.	Arsenic	British Anti Lewisite (BAL, dimercaprol) d-penicillamine, ferric hydroxide, sodium thiosulfate

S. No.	Poison	Antidotes
7.	Atropine	Pilocarpine, tannin, pyridostigmine
8.	Benzene Hexa Chloride	Barbiturates, chloral hydrate
9.	Barium	Sodium or magnesium sulfate.
10.	Barbiturates	Bemegride, stimulants
11.	Bromide	Sodium chloride
12.	Carbamates	Atropine
13.	Carbon tetrachloride	Calcium gluconate, glucose
14.	Copper	d-Penicillamine, potassium permanganate potassium ferrocyanide, molybdenum salts
15.	Coumarin anticoagulants	Vitamin K
16.	Curare	Neostigmine
17.	Cyanide	Sodium nitrite and sodium thiosulfate (i/v), amyl nitrate inhalation
18.	Diazepam	Flumazenil
19.	Dichloro Diphenyl Trichloroethane	Barbiturates, chloral hydrate (DDT)
20.	Ethylene glycol	Ethanol
21.	Fluorides	Calcium gluconate
22.	Heparin	Protamine sulphate
23.	Iron	Desferroxamine
24.	Lead	Calcium disodium EDTA, BAL
25.	Methyl alcohol	Ethanol
26.	Mercury	BAL, chloral hydrate
27.	Molybdenum	Copper
28.	Morphine	Nalorphine
29.	Nicotine	Caffeine, strychnine
30.	Nitrates & nitrites	Methylene blue solution intravenously
31.	Opioids	Naloxone
32.	Organophosphorus insecticides	2- PAM and Atropine
33.	Oxalates	Calcium gluconate
34.	Picrotoxin	Barbiturates

S. No.	Poison	Antidotes
35.	Phosphorus	Copper sulfate, potassium permanganate
36.	Snake envenomation	Antivenom
37.	Urea	5% acetic acid
38.	Xylazine	Tolazoline, yohimbine
39.	Zinc	Calcium disodium EDTA

*Refer Chapter 5 for drug doses and dosage intervals

CHAPTER 8

CONVERSION TABLES

WEIGHT EQUIVALENTS

1 pound (lb)	= 0.454 kg=454 grams=16 ounces (oz)
1 oz	= 28.35 gm
1 grain (gr.)	= 64.8 mg
1 kilogram (kg) = 2.2 pounds	= 1000 grams
1 gram	= 15.43 grains = 1000 mg
1 milligram	= 1000 microgram
1 microgram	= 1000 nanograms
1 ppm	= 1 µg/g=1 mg/Kg

VOLUME EQUIVALENTS

Household	Metric
1 drop (gt)	= 0.06 ml
15 drops (gtt)	= 1 ml
1 teaspoon (tsp)	= 5 ml
1 tablespoon (tbs)	= 15 ml
2 tablespoons	= 30 ml
1 ounce	= 30 ml
1 teacup	= 180 ml (6 oz)
1 cup	= 8 fl.oz = 237 ml = 16 tablespoons
1 glass	= 240 ml (8 oz)
1 measuring cup	= 240 ml (1/2 pint)
2 measuring cup	= 500 ml (1 pint)
1 gallon	= 4 qts = 8 pts = 128 fl.oz = 3.785 liters = 3785 ml
1 quart	= 2 pints = 32 fl. Oz = 946 ml
1 pint	= 2 cups = 16 fl.oz = 473 ml

PRESSURE EQUIVALENTS

1 cm H_2O	= 0.736 mm Hg	= 0.098 kPa
1 mm Hg (torr)	= 1.36 cm H_2O	= 0.133 kPa
1 kPa	= 7.5 mm Hg	= 10.2 cm H_2O
1 atm	= 760 mm Hg	= 1033.6 cm H_2O

TEMPERATURE CONVERSIONS

9 x °C	= (5 x °F) – 160
°C to °F	= (°C x 9/5) + 32 = °F
°F to °C	= (°F – 32) x 5/9 = °C

CATHETER, WIRE AND TUBING SIZE MEASUREMENTS

Gauge	Approximate external diameter (mm)	Approximate external diameter (in)	French Gauge*
29	0.330	0.013	1
22	0.711	0.028	2
18	1.270	0.050	3
17	1.473	0.058	4
16	1.651	0.065	5
14	2.108	0.083	6

***French gauge = 3 x external diameter in mm**

CONVERSION TABLES FOR WEIGHT TO KILOGRAMS TO BODY SURFACE AREA (M^2)

Dogs				Cats		Cattle	
Weight in Kg	m^2	Weight in Kg	m^2	Weight in Kg	m^2	Weight in Kg	m^2
0.5	0.06	29	0.95	2	0.159	1	0.10
1	0.10	30	0.97	2.5	0.184	5	0.29
2	0.16	31	1.00	3	0.208	10	0.46
3	0.21	32	1.02	3.5	0.231	20	0.72
4	0.25	33	1.04	4	0.252	50	1.32
5	0.29	34	1.06	4.5	0.273	100	2.10
6	0.33	35	1.08	5	0.292	250	3.83
7	0.37	36	1.10	5.5	0.311	450	5.64
8	0.40	37	1.12	6	0.330	500	6.04
9	0.44	38	1.14	6.5	0.348	800	8.24
10	0.47	39	1.16	7	0.366	1000	9.55
11	0.50	40	1.18	7.5	0.383		

Dogs				Cats		Cattle	
Weight in Kg	m^2	Weight in Kg	m^2	Weight in Kg	m^2	Weight in Kg	m^2
12	0.53	41	1.20	8	0.400		
13	0.56	42	1.22	8.5	0.416		
14	0.59	43	1.24	9	0.432		
15	0.61	44	1.26	9.5	0.449		
16	0.64	45	1.28	10	0.464		
17	0.67	46	1.30				
18	0.69	47	1.32				
19	0.72	48	1.33				
20	0.74	49	1.35				
21	0.77	50	1.37				
22	0.79	54	1.44				
23	0.82	58	1.51				
24	0.84	62	1.58				
25	0.85	66	1.65				
26	0.89	70	1.72				
27	0.90	74	1.78				
28	0.93	80	1.88				

BSA in m^2= ($K \times W^{2/3}$) x 10^{4-}

Where,

m^2 = square meter

BSA = body surface area

W = weight in grams

K = constant (dogs- 10.1, cats – 10.0)

CHAPTER 9

PROFORMAS

VETERINARY WOUND CERTIFICATE

No. ……………………….….

Date………………………….

This is to certify that at the request of (1) ………………………………………… I have this day examined (2) …………………………having the following identification marks………………… belonging to (3) ……………………………………………….. The said animal has got the following injuries on its body (4) ………………………………….. ……………………………………………………………………………I am of opinion that………………………………………………………………………………………… ...……………………………………………………………………………………………… …..

Place : **Signature of the Veterinarian**

Date : Qualifications

Registration No

Designation

INSTRUCTIONS

1. Name of the party, police officer or magistrate who has sent the animal. If from a police officer or magistrate, the number and date of the letter should be given.
2. The species, sex and colour of the animal examined, with full identification marks.
3. If received through a constable, his number and the station to which he belongs; if through messenger (in private cases) the name of the messenger or name and address of the owner.
4. The nature of the injuries with exact measurement and position. Kind of injury that is whether it is a permanent injury or can be cured in a definite period under treatment.

SOUNDNESS CERTIFICATE

This is to certify that I have this day (Date__________________) examined the horse a_________________________________, aged about___________________years, belonging to____residing at ____; has got the following identification marks.

1.	Color	:	
2.	Height	:	
3.	Natural markings	:	
4.	Acquired Markings	:	
5.	Branding	:	
6.	Congenital peculiarities	:	
7.	Whorls	:	

Opinion :

Place :

Date **:** **(Signature of the Veterinarian)**

Seal **Name and Designation**

PROFORMA FOR CERTIFICATE OF FITNESS TO TRAVEL DOGS / CATS

This Certificate should be completed and signed by a qualified Veterinary Surgeon

Date and time of examination	:	
Species of dogs/cats	:	
Number of cages	:	
Number of dogs/cats	:	
Sex	:	
Age	:	
Breed and identification marks, if any	:	
Transported from	:	………… to ………… Via

I hereby certify that I have read rules of the Transport of Animals Rules, 1978.

1. That, at the request of (consignor)………… I have examined the above mentioned dogs/cats in their travelling cages not more than 12 hours before their departure.
2. That each of the dogs/cats appeared to be in good health, free from signs of injury, contagious and infectious disease including rabies and in a fit condition to travel by rail/road/inland/ waterway/ sea/air.
3. That the dogs/cats were adequately fed and watered for the purpose of the jou rney.
4. That the dogs/cats have been vaccinated.
   (a) Type of vaccine/s:
   (b) Date of vaccination/s:

Signed:…………… …

Address:………………

……………………………

Date:……………

Qualifications

PROFORMA FOR CERTIFICATE OF FITNESS TO TRAVEL - CATTLE

This Certificate should be completed and signed by a qualified Veterinary Surgeon

Date and time of examination	:	
Species of cattle	:	
Number of trucks/railway wagons	:	
Number of cattle	:	
Sex	:	
Age	:	
Breed and identification marks, if any	:	
Transported from	:	………… to ………… Via

I hereby certify that I have read rules of the Transport of Animals Rules, 1978.

1. That, at the request of (consignor)……………..… I have examined the above mentioned cattle in the goods vehicle/railway wagons not more than 12 hours before their departure.
2. That each cattle appeared to be in a fit condition to travel by rail/road and is not showing any signs of infectious or contagious or parasitic disease(s).
3. That the cattle were adequately fed and watered for the purpose of the journey.
4. That the cattle have been vaccinated.
   (a) Type of vaccine/s:
   (b) Date of vaccination/s:

Signed:…………… …

Address:………………

…………………………

Date:……………

Qualifications

PROFORMA FOR CERTIFICATE OF FITNESS TO TRAVEL – SHEEP & GOATS

This Certificate should be completed and signed by a qualified Veterinary Surgeon

Date and time of examination	:	
Species of animals	:	
Number of animals	:	
Sex	:	
Age	:	

I hereby certify that I have read rules of the Transport of Animals Rules, 1978.

1. That, at the request of (consignor)………… I have examined the above mentioned animals in their travelling cages not more than 12 hours before their departure.
2. That each of the animals appeared to be in a fit condition to travel by rail/road and is not showing any signs of infectious or contagious or parasitic disease(s) and that it has been vaccinated against any infectious or contagious or parasitic disease(s).
3. That the animals were adequately fed and watered for the purpose of the journey.
4. That the animals have been vaccinated.

   (a) Type of vaccine/s:

   (b) Date of vaccination/s:

Signed:…………… …

Address:………………

……………………………

Date:……………

Qualifications

PROFORMA FOR CERTIFICATE OF FITNESS TO TRAVEL - EQUINES

This Certificate should be completed and signed by a qualified Veterinary Surgeon

Date and time of examination	:	
Species of Equines	:	
Number of Equines	:	
Sex	:	
Age	:	
Breed and identification marks, if any	:	
Transported from	:	………… to ………… Via

I hereby certify that I have read rules of the Transport of Animals Rules, 1978.

1. That, at the request of (consignor)………… I have examined the above mentioned equines not more than 12 hours before their departure.
2. That each equines appeared to be in a fit condition to travel by rail/road/sea and is not showing any signs of any infectious or contagious disease(s) and that it has been vaccinated against any infectious or contagious disease(s).
3. That the equines were adequately fed and watered for the purpose of the journey.
4. That the equines have been vaccinated.

(a) Type of vaccine/s:

(b) Date of vaccination/s:

Signed:…………… …

Address:………………

…………………………

Date:……………

Qualifications

CHAPTER 10

VARIOUS FIELD TESTS IN VETERINARY PRACTICE

S. No.	Content
1.	Hemoglobin estimation – sahil’s acid haematin method
2.	Famacha chart
3.	Examination of blood
4.	Saline agglutination test
5.	Buffy coat examination
6.	Field’s stain
7.	Glucometer
8.	Benedict's test
9.	Rothera's test
10.	Benzidine test
11.	Rivalta test
12.	Glutaraldehyde test
13.	Refractometer for colostrum
14.	Abomasocentesis
15.	Thoracocentesis
16.	Pericardiocentesis
17.	Abdominocentesis
18.	Liver biopsy
19.	Fine needle aspiration biopsy
20.	Rumen fluid examination
21.	Modified diphenylamine test
22.	Picrate paper test
23.	California mastitis test
24.	Cystocentesis
25.	Urethral catheterization
26.	Diascopy
27.	Wood’s lamp examination
28.	Skin biopsy
29.	Examination of skin scrappings
30.	Examination of nasal discharge
31.	Acetate tape impression
32.	Schirmer tear test

HAEMOGLOBIN ESTIMATION - SAHH's ACID HAEMATIN METHOD

Blood is treated with 0.1 N HCl → acid separates globin from haematin → formation of acid hematin (brown colour) → colour matched with standards. The concentration of Hb is read directly.

Procedure

1. 0.1 N HCl is added in diluting tube of hemoglobinometer upto graduation 2
2. 0.02 ml blood is taken in the pipette and mixed with the HCl
3. Leave undisturbed for 10 minutes
4. Keep the hemoglobinometer tube in the comparator box and add either distilled water or 0.1N HCl drop by drop stirring with the glass rod till it's color matches with the comparator glass
5. Remove the stirrer and take reading by noting the height of diluted acid hematin
6. Result expressed in g%.

Sahil's Haemoglobinometer

FAMACHA CHART

Indications

- To estimate the level of anemia in sheep and goats
- The "FAMACHA" chart assigns a number from 1 to 5 to each level of colour in the eyelid which ranges from Red to reddish pink, to pink, to pinkish-white and to white.

Procedure

Expose the lower eye mucous membranes and matching them to the equivalent color on the FAMACHA card

↓

Match the color of the pinkest portion of the mucous membranes to the FAMACHA card

Interpretation

Category 4 & 5 - Deworm sheep & goats

Category 1 & 2 - Don't deworm unless there is other evidence of parasitic diseases

Category 3: Consider deworming if,

- More than 10% of flock/herd scores a 4 or 5
- Lambs and kids
- Pregnant or lactating ewes/does

Estimation of level of aemia by using Famacha Chart in a Goat

EXAMINATION OF BLOOD

Blood is examined to detect the blood parasites, like *Trypanosoma, Babesia, Theileria* and microfilariae and rickettsial organisms like *Anaplasma and Ehrlichia*. Blood is taken from either ear vein or tip of the ear of the animal and from wing vein in case of bird. The blood can be examined either directly or by preparing blood smears.

Direct or wet blood film examination

- It is used for the detection of motile parasites like microfilariae and *Trypanosoma*.

Place a drop of blood on a clean glass slide and covered with a coverslip

↓

Examine it under for microfilariae in low power for microfilariae and *Trypanosoma* in high power

↓

Microfilariae have a serpentine movement, lashing forward while the trypanosomes have a circular or rotary motion.

Blood smear examination

Thin blood smear preparation

- Thin blood smear is prepared in heavy infection.

Take a small drop of blood in a clean, grease free glass slide

↓

Place a spreader slide at 30-45° angle in front of the blood drop

↓

Push the spreader slide to touch the blood and allow it to spread along its edges

↓

Draw the spreader slide smoothly and steadily along the surface of examination from slide up to the far end

↓

Tongue shaped thin smear is examined after drying and staining

Thick blood smear preparation

- Thick blood smear is prepared in case of low parasitic infection.

Large blood drop is taken at the centre of a clean glass slide

↓

Blood is spreaded in circular motion by using corner of another slide or glass rod in 2 cm diameter area

↓

Air dry the thick blood smear

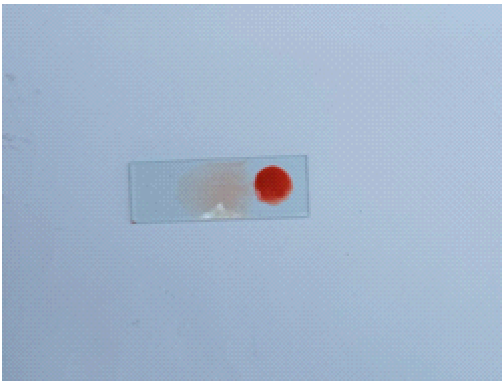
↓

Blood smear is kept under distilled water for dehaemoglobination

↓

Thick smear is examined after drying and staining

Note: If the smear is to be stored for more than a day or so before staining, it should be fixed with absolute methyl alcohol before storage.



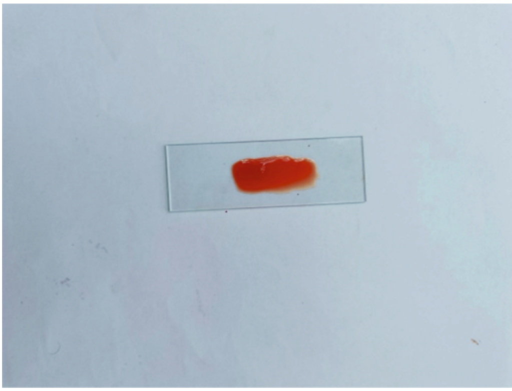
Thick and thin blood smear preparation procedure

SALINE AGGLUTINATION TEST

- Used to determine the red cell agglutinates
- Presence of red cell agglutinate indicates that anaemia is due to immune-mediated haemolysis

PROCEDURE

- Few drops of saline is added in a glass slide
- Small amount of EDTA blood is added on to the surface of the saline
- Mix until formation of slight pink colour by gently rocking the slide
- Place a coverslip and examine it under low power microscope
- Negative: Red cells are freely floating apart from each other
- Positive: Formation of clumps / bunches of grapes of red cells (rouleaux formation)



Saline agglutination test procedure

BUFFY COAT EXAMINATION

- The buffy coat is the concentration of all the white blood cells and platelets in a blood sample

Indications

- Detection of infection with malaria or other blood parasites
- *Leishmania donovani*
- Trypanosomes
- Microfilariae of *Dirofilaria immitis* in dogs
- *Histoplasma capsulatum*
- Mast cell tumor

Procedure

- Microcapillary tube or wintrobe tube is filled with EDTA-treated anticoagulated blood
- Blood sample is centrifuged for 15 minutes at 1000 rpm
- Supernatant containing the plasma above the buffy coat layer is removed
- The buffy coat layer, along with red cells is transferred to a slide and mixed well.
- Thin smear is prepared and examined after staining.

FIELD STAIN

- Field stain contain methylene blue and eosin
- Basic and acidic dyes induce several colours when applied to cells.

Procedure

- Take the slide with smear
- Fix it in methanol for 30 seconds
- Dip the slide in Field's stain B for 5-6 seconds
- Wash the slide in running tap water
- Dip the slide in Field's stain A for 20-30 seconds
- Wash the slide in running tap water
- Air dry the slide and observe under oil immersion objective lens

Field stains

GLUCOMETER

- Used to measure blood sugar levels instantly in healthy animals.
- Glucometer uses electrochemical test strips to perform the measurement which is based on electrochemical technology.

Test strip contains the enzyme glucose oxidase → reacts with the glucose in the blood sample → formation of gluconic acid → reacts with ferricyanide in testing strip → ferricyanide and gluconic acid combine together to form ferrocyanide → electronic current runs through the blood sample on the strip → read ferrocyanide → determine glucose

Glucometer set

BENEDICT'S TEST

- To estimate glucose in the urine sample

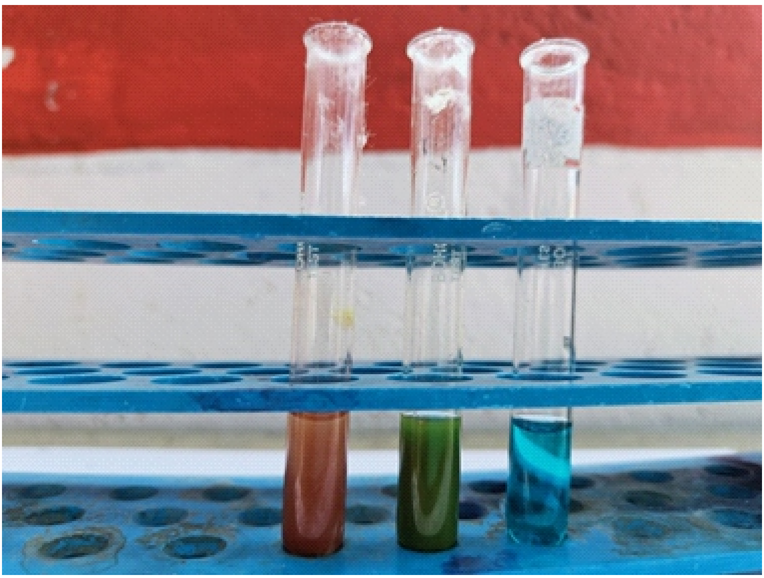
Benedict's reagent

- **Solution 1:** Cupric sulphate crystals – 17.3 grams, distilled water – 100 ml
- **Solution 2:** Sodium carbonate – 100 grams, sodium citrate – 173 grams, distilled water – 700 ml
- Cool the solution 2 to room temperature and pour into the solution 1 by stirring constantly and make up the volume to 1000 ml with distilled water.

Procedure

5 ml of Benedict's reagent + 8 to 10 drops of urine → boil the mixture & cool down → observe the colour change

- Blue: sugar absent
- Green: +
- Yellow: ++
- Orange: +++
- Brick red: ++++



Benedict's test reactions: Positive for glucose

ROTHERA'S TEST

To estimate ketone bodies in the urine

- Rothera's powder: Sodium nitroprusside = 0.75 gm and Ammonium sulphate = 20gm (Mix and pulverize).

Procedure

- Take little amount of ammonium sulphate in a test tube
- Add 5 ml of urine in it and mix well
- Add one or two drops of 5% sodium nitroprusside solution (freshly prepared) and mix well
- Add 1-2 ml of concentrated ammonium hydroxide
- Observe the pink-purple ring at the interface
- Positive: Formation of purple permanganate-colored ring
- Negative: Absence of purple permanganate-colored ring formation

Rothera's test: Formation of purple permanganate colored ring (Positive for ketone bodies)

BENZIDINE TEST

Used to determine the blood in urine.

- Benzidine reagent: Benzidine dihydrochloride, Glacial acetic acid and 3% hydrogen peroxide

PROCEDURE

- Take 3 ml of fresh benzidine reagent in a test tube → Add 2 ml of urine → Add 1 ml of 3% hydrogen peroxide → Mix and observe for colour change → Presence of hemoglobin in urine: Presence of blue or green colour

SULKOWITCH TEST

Used to determine calcium in urine.

Sulkowitch reagent:

- Oxalic acid: 2.5 grams
- Ammonium oxalate crystals: 2.5 grams
- Distilled water: 45 ml
- Add glacial acetic acid: 5 ml
- Take equal parts of urine and reagent and observe for precipitation

Result

S. No.	Precipitation	Grades	Serum calcium level
1.	No	-	< 5mg%
2.	Mild	+	5 – 10 mg%
3.	Cloudy	++	10 – 15 mg%
4.	Milky white	+++	15 – 20 mg%
5.	Heavy chalky	++++	20 mg% and above

RIVALTA TEST

Used to differentiate transudate and exudate. Used in Feline Infectious Peritionitis.

PROCEDURE

- 10 ml glass tube is filled with 7–8 ml of distilled water
- 1 drop of acetic acid (98%) is added and mixed thoroughly.
- On the surface 1 drop of the effusion fluid is added
- If the drop dissipates, the test is negative, indicating a transudate.
- If the drop precipitates, the test is positive, indicating an exudate.

GLUTARALDEHYDE TEST

- Used to evaluate visual and semi quantitative identification of an inflammatory process in cattle

PROCEDURE

- 50 mL of 25% glutaraldehyde, 1 g of Na2 -EDTA and 1 liter of 0.9% NaCl solution was used to for prepare the glutaraldehyde test solution
- Collect 2 mL of blood and mix with 2 ml of glutaraldehyde test solution in an anticoagulant collection tube
- Measured for up to 15 minutes or until coagulation

Glutaraldehyde Category	Coagulation Time	Interpretation	Inflammatory process
1	0 – ≤ 3 minutes	High increase in concentration of fibrinogen and / or immunoglobulin	Severe
2	>3 – ≤ 6 minutes	Moderate increase in concentration of fibrinogen and/or immunoglobulin	Moderate
3	>6 – ≤15 minutes	Low increase in concentration of fibrinogen and/or immunoglobulin	Mild
4	> 15 minutes	No increase in concentration of fibrinogen and/or immunoglobulin	Not detectable

REFRACTOMETER FOR COLOSTRUM

- Brix refractometer is used to measure IgG immunoglobulin in the colostrum. The scale in a Brix refractometer is designed to measure the amount of sucrose in a solution, but Brix values can be related to IgG in colostrum.
- Concentration of IgG in colostrum will vary from 20 to over 100 mg/ml. The difference between 20 and 100 mg/ml of IgG in colostrum indicates the difference between failure and success in passive transfer of immunity.
- Colostrum with a Brix value ≥ 22% should be used
- Colostrum with a Brix value of < 18% should be discarded
- Colostrum has a Brix value of ≥ 18% but < 22% supplementation should be considered

PROCEDURE

- Calibration of the refractometer should be checked.
- In properly calibrated instrument distilled water produce a reading of zero
- Few drops of colostrum are placed on the prism and the sample cover is lowered
- Refractometer is held up to a light source and held perpendicular to the light
- Read the brix value at the line between the light and dark areas that appear on the scale.

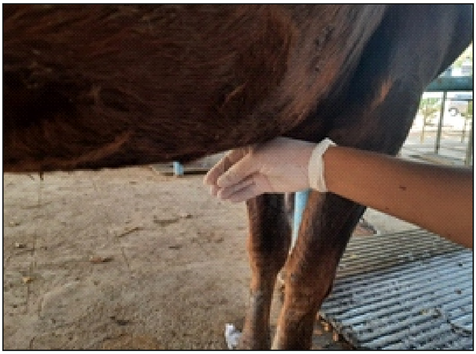
Refractometer

ABOMASOCENTESIS

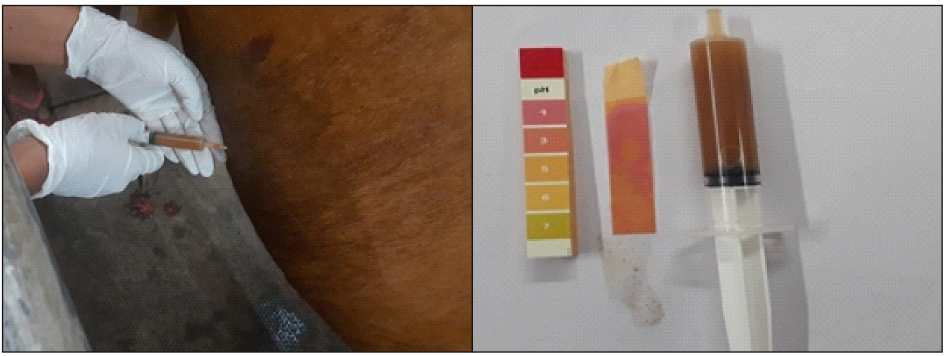
- For the collection of fluid from abomasum

Location: Between sternum and umbilicus slightly towards the right side to the middle line. Use 16 gauge needle about 8-10 cm long. It should be differentiated from ruminal fluid (olive green in color) whereas abomasal fluid (pH 2 to 4) has characteristic sour odour distinct from that of rumen ingesta.

LIPTAK TEST- in abomasal displacement ping area is punctured and the pH of 2 - 4 and absence of protozoa are observed.



Site for abomasocentesis



Liptak test: Right displacement of abomasum

THORACOCENTESIS

Indications: This is of value in the presence of fluid/air in the pleural sac.

E.g., exudative type of pleurisy, hydrothorax in CHF, pyothorax, haemothorax, chylothorax and pneumothorax.

Large animals: 18G needle is introduced between 6th and 7th intercostal space below the anticipated level of the fluid line and aspiration is done with syringe.

Dog and cat: 20G or 21G butterfly i/v needle/scalp vein set with three way stop cock used. Needle is inserted between 7th and 8th intercostal space.

Care must be taken to avoid puncture of pericardial sac unless a sample of pericardial fluid is specifically required. Examination of fluid is helpful for diagnosing the severity of disease, to observe for the presence of microbes and neoplastic cells.

Cattle: Site for thoracocentesis

PERICARDIOCENTESIS IN CATTLE

Indications: Pericardial effusion.

Site: Between the 4th and 6th intercostal space of the left hemithorax, slightly below the costochondral junction. (OR) One hands breadth above the elbow joint in left side between 3rd to 5 th rib.

Pericardial drainage catheter fixation in cattle with pericardial effusion

ABDOMINOCENTESIS

Indications

- Ascites
- Inflammatory exudation

Complications

- Perforation of a hollow viscus
- Laceration of abdominal organs (intestine, spleen)
- Peritonitis (iatrogenic)

The fluid is collected in tube having EDTA for cytologic examination, cell count and specific gravity estimation. Fluoride oxalate is used for glucose and lactate estimation.

Abdominocentesis sites

Cattle

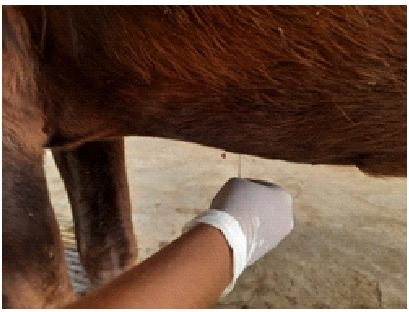
- Ventral anterior abdomen: midway between the xiphisternum and the umbilicus in the midline
- Ventral anterior abdomen: 8-10 cm caudal to the xiphisternum and 8-10 cm to the left or right of the midline
- Posterior abdomen: anterior to the attachment of the mammary gland to the body wall.

Dogs

Slightly caudal and lateral to the umbilicus.

Horse

Choose an area in the most dependent portion of the abdomen. Usually directly on the midline 5 cm caudal to the xiphoid.



Abdominocentesis in cattle:
Cranial site

Abdominocentesis in cattle:
Caudal site in peritoneal effusion

LIVER BIOPSY

Indications: Diffuse or zonal lesions due to toxic, infectious, metabolic liver diseases.
Instruments

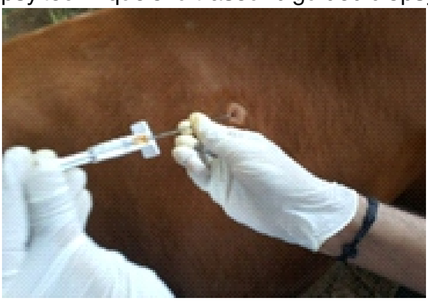
- Tru-cut biopsy needle
- Vim Silvermann biopsy needle

Cattle

- Site of the biopsy is 15 cm below the transverse processes in the 11th right intercostal space.
- Site is also defined by imaginary lines from the wing of the ilium to the point of the elbow and point of the shoulder.
- Hair is clipped and aseptically prepared.
- Local anaesthetic is infiltrated subcutaneously and more deeply into the intercostal muscles beneath.
- Stab incision is made through the skin at this site.
- The biopsy needle is pushed through the skin incision and aimed towards the opposite elbow. Needle is then pushed into the stroma of the liver.
- Ultrasonography can be used to guide the placement of the biopsy needle.
- Passage of the needle through the edge of the diaphragm and the liver gives a slight grating sensation. A biopsy is then taken and the needle withdrawn.
- Sample can then be placed in 10% formal-saline for histopathology, or the fresh sample can be used for specific gravity tests and chemical analysis for lipid content.
- Liver pathology such as fatty liver syndrome and a fibrotic liver from ragwort poisoning can be detected.

Dog

- Restrain the animal in dorsal recumbency.
- Blind percutaneous biopsy technique or ultrasound guided biopsy is to be followed.



Liver biopsy procedure

FINE NEEDLE ASPIRATION BIOPSY

Indications

- To obtain a sample of cells from an accessible mass for cytologic examination
- To help differentiate among inflammation, neoplasia and hyperplasia such as lymph nodes or mammary glands

Procedure

- Restrain the animal to allow exposure of mass/lymph node to be biopsied
- Sterilize the biopsy site
- Attach the needle firmly to the syringe
- Grasp the mass between the thumb and fingers of the free hand. Insert the needle through the skin and into the mass
- Apply negative pressure on the syringe (withdraw the plunger to the 3-5 ml mark) while the needle is still in the mass.
- Gently release and reapply this negative pressure three or four times.
- Separate the needle from the syringe
- Retract the syringe plunger to the 3 ml mark; then reattach the needle.
- Quickly eject the material in the needle onto a clean glass slide
- If the aspirated material is liquid, a push slide can be made; if the material is more viscous, a pull smear should be made.

Cattle: Fine needle aspiration biopsy

RUMEN FLUID EXAMINATION

Rumen fluid extraction pump (for cattle, sheep and goat) consists of:

- A specially designed suction strainer.
- Three meter nylon tube with perforation at the tip which is covered by a stainless steel spiral sound.
- A suction pump.
- An air tight sampling bottle with two way "T" joint.

Limitation

If the animal has been starving for considerable time and the rumen content is semisolid in nature, rumen fluid may not be obtained by using the rumen fluid extraction pump. In such cases, physiological saline is to be pumped in and massaged well before collection.

Important points to be observed before and after collection of rumen fluid

- About 200 ml of rumen fluid sample is to be collected for various tests.
- The rumen fluid can be kept in room temperature for 9 hours and in refrigerator for 24 hrs.
- After collection, rumen fluid container should be immediately closed air tight.
- During collection discard the first collection of rumen fluid and then collect the sample so that error in pH can be minimized.

After use

For longevity and efficient usage of this instrument, clean the suction strainer, suction line and

Common tests conducted in rumen liquor

Colour	Redox potential or Methylene blue reduction test (MBRT)
Odour	Volatile Fatty acid
Consistency	Total Acidity (Titratable acidity)
Sedimentation activity time (SAT) or Stratification test	Chloride content
pH	Protozoa
Cellulose digestion test	Bacteria
Glucose fermentation test	

Examination of rumen fluid sample

Take 100 ml of rumen fluid in a beaker and conduct the following bio-physical examination in well lighted area.

A. COLOUR

Normal colours	
Pure green	Grazing
Yellowish brown	Straw feeding
Grey/Brownish green	Concentrate and straw feeding
Abnormal colours	
Dark brown/Dark green	Simple inactivity of flora and fauna, rumen acidosis
Slightly milky white	Chronic rumen acidosis
Milky green	Acute rumen acidosis
Dark green	Hydrochloric acidosis
Green black	Vagus indigestion, decomposition of food

B. ODOUR

Normal	Aromatic
Abnormal	
Stale/indifferent	Inactive rumen juice
Acrid (acid) acidosis	Lactacidosis/ hydrochloric acidosis, pyloric obstruction
Foul/Putrid	Protein overfeeding
Slightly ammoniacal	Rumen alkalosis (urea poisoning)
Musty /faecal	Vagus indigestion

C. CONSISTENCY

Normal	Slightly viscous
Extremely viscous	Saliva mixed
Watery	Inactive rumen fluid
Foaming	Abomasal dilatation or frothy bloat
Mixture of watery and foam	Rumen decomposition (E.coil, Proteus)
Slimy pulp	Overfeeding
Semiliquid	Vagus indigestion

M. PROTOZOA

Both ciliates and flagellates are present in rumen. But only ciliates are of physiological importance in virtue of their numbers and mass. The majority of ruminal protozoa belong to the family Ophryoscolecidae. Their numbers and size distribution provide information on protozoal activity within the forestomach.

Place a drop of rumen fluid on a slide and cover with a cover glass and examine under lower power (80 or 100 magnification).

Assess the following characters

1. Density

No. of protozoa	Motility	Grade
Unable to count (More no. of protozoa)	Vigorous	++++
More than 30 protozoa/low power field (LPF)	Abundant	+++
10 - 30 protozoa/LPF	Moderate	++
1 - 10 protozoa/LPF	Few	+
No protozoa	None	-

More the density of protozoa, more active is the rumen fluid

- Simple inactivity of flora and fauna -/+
- Rumen decomposition/alkalosis -/+
- Acute rumen acidosis -

2. Proportion of large, medium and small infusoria

- Large and medium ciliates in large numbers - Active rumen fluid
- Only small ciliates - Fore stomach indigestion

3. Proportion of dead to live protozoa

- Severe digestive disorders like rumen acidosis (pH less than 5) - Entire microfauna dead.
- Moderate digestive disorders - Proportion of dead to live protozoa increases.

4. Iodophilic activity

- Add a drop of lugol's Iodine to 2-3 drops of rumen fluid on a glass slide and place a cover slip on it and examine under the microscope.
- Iodophilic activity of protozoa can be recognized by black colouration of starch contents in protozoa.
- It can be graded as 0, +, ++, +++ depending on quantity of starch contained.

Rumen fluid extraction pump – Foot operated

MODIFIED DIPHENYLAMINE TEST

Used to diagnose nitrate toxicity

- Put 1 or 2 drops of plasma/ urine/ rumen fluid on glass slide
- Add a drop of test solution (0. 5g of diphenylamine, 20 ml distilled water and concentrated sulphuric acid sufficient to make 100 ml) and mix well.
- An intense blue colour within 10 seconds indicates a concentration of greater than 1% nitrate which is more than the permissible level.

PICRATE PAPER TEST

Used to detect hydrocyanic acid toxicity

- Reagent papers are prepared by mixing 0.5g of picric acid and 5 g sodium carbonate in 100 ml of water.
- Filter papers are dipped in the reagent and allowed to dry in a dark place.
- A drop of rumen fluid is placed on the test paper.
- Red discoloration is a positive reaction.
- The test is designed only to detect free hydrocyanic acid.
- It may not be positive when cyanides are present in different forms.

CALIFORNIA MASTITIS TEST

The California Mastitis Test (CMT) is a diagnostic tool to aid in the quick diagnosis of sub clinical mastitis in dairy cows and for an udder health management program.

Indications

- The CMT will only trigger a visible reaction with a concentration of 4,00,000 cells/ml or more.
- The degree of gelling indicates the presence and severity of mastitis.
- The change in colour indicates the pH variation of the milk and therefore, the level of inflammation.
- Cows in early and late stages of lactation may give false positive reactions.

CMT reagent

- **Solution A:** Sodium lauryl sulphate 30 gram is dissolved in 1000 ml of distilled water ⟶ heat to 50°C to make clear solution.
- **Solution B:** Bromocresol purple 0.5 gram is dissolved in 100 ml of dissolved water.
- Mix 100 ml of solution A to 2 ml of solution B to make CMT reagent

Procedure

Prepare the cow for milking as per SOP

↓

After discarding the first stream of milk, draw 2 – 3 ml of milk into the shallow cups on the paddle, keeping the quarters separate

↓

Drain excess milk

(The ideal amount of milk is that which remains in the cup when the paddle is tilted to an almost vertical position)

↓

Add an equal amount of the CMT reagent solution

↓

Mix the reagent and milk

(Gently rotate the paddle in horizontal plane, swirling the mixture for 10-30 seconds)

↓

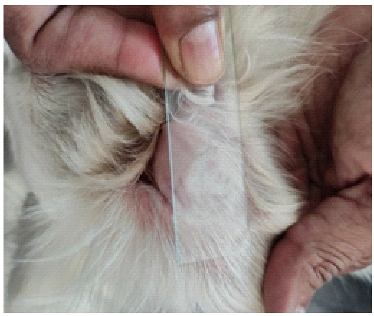
Formation of gel indicates positive reaction

California mastitis test: Gel formation indicates positive

DIASCOPY

- Diascopy is performed by pressing a clear piece of plastic or glass over an erythematous lesion.
- If the lesion blanches on pressure, the reddish color is due to vascular engorgement (Positive diascopy).
- If it does not, there is haemorrhage into the skin (petechial or ecchymosis) (Negative diascopy).

Negative diascopy



Positive diascopy

WOOD'S LAMP EXAMINATION

- The wood's lamp is an ultraviolet light with a light wave of 235.7 nm that is filtered through a cobalt or nickel filter.
- Turn on the wood's lamp and allowed to warm up for 5 to 10 minutes.
- Place the animal in a dark room and expose the light on skin for 3-5 minutes
- *M. canis* result in a yellow green fluorescence in 30 – 80% of the isolates
- Fluorescence is not present in scales or crusts or in cultures of dermatophytes

Woods lamp

SKIN BIOPSY

- Biopsies are often performed with local anaesthesia with 2% lidocaine.
- In general a 6 mm biopsy punch provides an adequate specimen. 4 mm punches are reserved for near the eye, on the pinna and on the nasal planum and foot pads.
- It is imperative not to include any significant amount of normal skin margin with punch biopsy specimens.
- Lesions should be centered in the biopsy specimen.

Procedure

- Physically restrain the animal and infiltrate local anaesthesia
- Sites are gently clipped (if needed)
- Careful in not to remove surface keratin and the surface is left untouched or gently cleaned by daubing or soaking with a solution of 70% alcohol. Sites should not be prepared with other antiseptics (eg. iodophors)
- Under no circumstances should biopsy sites be scrubbed.
- 1 to 2 ml of local anaesthetic is injected subcutaneously through 25 gauge needle.
- When the disease of fat is suspected, in which case ring blocks or regional or general anaesthesia should be used, because injection into fat distorts the tissues.
- The desired lesion is punched or excised, including the underlying fat.
- Skin biopsy specimen should be gently blotted to remove artifactual blood and fixed in 10% formalin.

Punch biopsy forceps

EXAMINATION OF SKIN SCRAPINGS

- Skin scrapings are examined to diagnose mite infestations.
- The periphery of lesion is richer source of infection than the center.

Dampen the area to be scrapped with water or physiological saline or dip the scalpel blade in mineral oil

↓

Squeeze the lesion and scrape deeply enough with a blunt scalpel to produce pin point haemorrhage

↓

Direct examination

Place a small pinch of scrapings on glass slide → add a drop of mineral oil → put coverslip and examine it under low power microscope

Sedimentation method

Scrapings are taken in a test tube and 10% sodium or potassium hydroxide (5 ml) is added

↓

Boiled for 10 minutes or left for 12-24 hours

↓

Centrifuged for 2 minutes at 1000 rpm

↓

Sediment is examined under microscope in low power

Skin scraping procedure

ACETATE TAPE IMPRESSION

- Used to find superficial ectoparasites such as Cheyletiella mites, poultry mites and cat fur mites.
- Clear, pressure sensitive acetate tape is pressed to the hair surface and to the skin adjacent to parted hairs or in shaved areas.
- Superficial scales and debris are collected.
- The tape is then stuck with pressure on a microscope slide and examined.

Acetate tape impression procedure

EXAMINATION OF NASAL DISCHARGE

- Used to examine *Shistosoma nasalis* egg in the diagnosis of nasal granuloma.

Collect either the nasal discharge or scrape the growth in the nasal cavity

↓

Samples are taken in a test tube and 10% sodium or potassium hydroxide (5 ml) is added

↓

Boiled for 10 minutes or left for 12-24 hours

↓

Centrifuged for 2 minutes at 1000 rpm

↓

Sediment is examined under microscope in low power for eggs

SCHIRMER TEAR TEST

- Used to evaluates tear production in millimeters of wetting on a paper strip.
- The strip is placed on the ventral conjunctival fornix of both eyes and left for 1 minute
- Remove and immediately place on measuring gauge to determine the milliliters of wetting per minute
- Diminished production of aqueous portion of tear (keratoconjunctivitis sicca)
  - Cat – <5 mm
  - Other domestic animals – < 10 mm

Schirmer tear test in a normal dog

CHAPTER 11

ANIMAL REPRODUCTION

1. The duration of proestrus, estrus, metestrus and diestrus in cow **is 3 days, 12-24 hrs, 3-5 days and 13 days**
2. The duration of proestrus, estrus, metestrus and diestrus in mare is **3 days, 4-7 days, 3-5 days and 6-10 days**
3. The duration of proestrus, estrus, metestrus and diestrus in sow is **3 days, 2-4 days, 3-4 days and 9-13 days**
4. The duration of proestrus, estrus, metestrus and diestrus in ewe is **2 days, 1-2 days, 3-5 days and 7-10 days**
5. In bitches the duration of proestrus is **(3-16 days)**and the duration of estrus is **(4-12days)**
6. **Monoestrus animals:** Those animals which are having only one estrous cycle per year. Ex: wild animals
7. **Polyestrus animals:** Those animals which are having frequent periodic oestrus cycles throughout the year. Ex: cow, sow and mare
8. **Sheep and Mare** - Seasonally poly oestrus animal
9. **Mare** - Long day breeder
10. **Sheep** – Short day breeder
11. The bitch are more nearly monoestrus animals
12. However bitches may have periods of oestrus each year
13. Average recommended age for first service in mare is **2-3 years**
14. Average recommended age for first service in cow is **14-22 months**
15. Average recommended age for first service in ewe is **12-18 months**
16. Average recommended age for first service in doe is **12-18 months**
17. Average recommended age for first service in sow is **8-9 months**
18. Average recommended age for first service in bitch is **12-18 months**
19. Time of ovulation in cow is **12-16 hours after the end of oestrum.**
20. Time of ovulation in mare is **1-2 days before the end of oestrum.**
21. Time of ovulation in ewe is **12-24 hours before the end of oestrum.**
22. Time of ovulation in doe is **last day of oestrum or 24 hours before the end of oestrum.**
23. Time of ovulation in sow is **30-40 hours after the onset of oestrum.**
24. Time of ovulation in bitch is **1-2 days after onset of true oestrum.**
25. Time of ovulation in cat is **24-30 hours after the coitus or copulation.**

26. Optimum time for breeding or services or AI

Mare	**2-4 days before the end of oestrum or 2nd or 3rdof oestrus**
Cow	**Mid to late oestrum**
Ewe	**18-24 hours after the onset of oestrum**
Doe	**24-36 hours after the onset of oestrum**
Sow	**12-30 hours after the onset of oestrum**
Bitch	**2-3 days after the onset of true oestrum**

27. The transport of fertilized ovum from oviduct to uterus occurs at **4-56days** in mare

28. The transport of fertilized ovum from oviduct to uterus occurs at **3-4 days** in cows, ewes and does

29. The transport of fertilized ovum from oviduct to uterus occurs at **2-3 days** in sow

30. The transport of fertilized ovum from oviduct to uterus occurs at **6-8 days** in bitches

31. The transport of fertilized ovum from oviduct to uterus occurs at **4-8 days** in cat

32. Advisable time for breeding / AI after parturition

Mare	**25-35 days postpartum or second estrum**
Cow	**60-90 days after parturition**
Goat	**60-90 days after parturition**
Sow	**First postpartum oestrum / 4-9 days after weaning of piglets,**
Dog & Cat	**First postpartum oestrum or 3-4 weeks after weaning their young ones**

33. Duration of Oestrous cycle

Animals	Duration	Average
Mare	**19-23 days**	**21 days**
Cow	**18-24 days**	**21 days**
Ewe	**14-20 days**	**16.5 days**
Goat	**15-24 days**	**20 days**
Sow	**18-24 days**	**21 days**
Dog	**1-3 cycles / year**	**2 cycles / year**
Cat	**15-4 times a year**	

33. Age at puberty

Animal	Months
Mare	**10-24 (Avg:18 months)**
Cow	**6-18 months**
Buffalo	**12-24 months**
Ewe	**12-18 months**
Goat	**12-18 months**
Sow	**8-9 months**
Dog	**6-12 months**
Cat	**6-12 months**

33. Duration of pregnancy

Species	Duration (in days)	Average (in days)
Cow	**273 – 296**	**278**
Mare	**327- 357**	**335**
Buffalo	**305 ± 5**	**305**
Sheep	**140 – 155**	**148**
Goat	**148-156**	**150**
Swine	**111- 116**	**114**
Dog	**60-63**	**63**
Cat	**56- 65**	
Rabbit	**30-32**	
Indian elephant	**615- 650**	
Deer	**200-210**	
Lion	**105-112**	
Tiger	**105-113**	
Camel (Dromedary)	**315- 350**	

33. Stages of labour

First stage of labour

Species	Duration(in hours)
Mare	**1-4 hrs**
Cow	**2-6 hrs**
Buffalo	**2-6 hrs**
Sow	**2-12 hrs**
Dog	**6-12 hrs**
Cat	**6-12 hrs**

Second stage labour

Species		Duration
Mare		**5- 40 min (avg. 20 min)**
Cow	Pleuriparous	**0.5-1 hrs**
	Primiparous	**0.5- 3 hrs**
Sheep and goat		**0.5-2 hrs**
Sow		**2.5- 3 hrs (range.1-5 hrs)**
Dog		**3-6 hrs(up to 24 hrs)**
Cat		**3-6 hrs (up to 12 hrs)**

Third stage labour

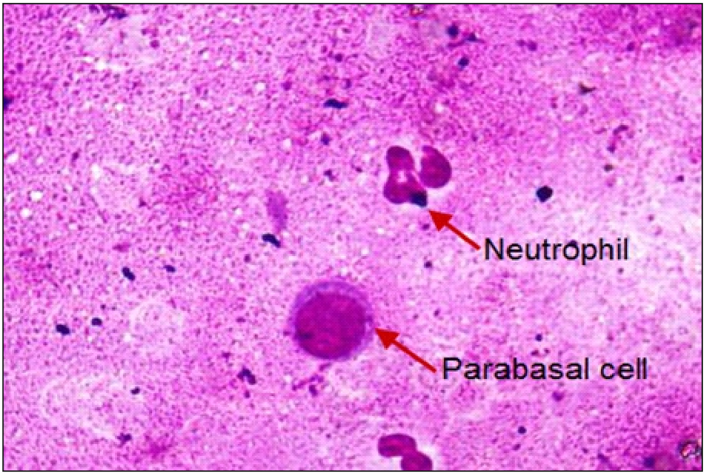
Species	Duration
Mare	**0.5-3 hrs**
Cow	**0.5-8 hrs (upto 12 hrs)**
Buffalo	**0.5-8 hrs (upto 12 hrs)**
Sheep and goat	**0.5 – 8hrs**
Sow	**1-4 hrs**

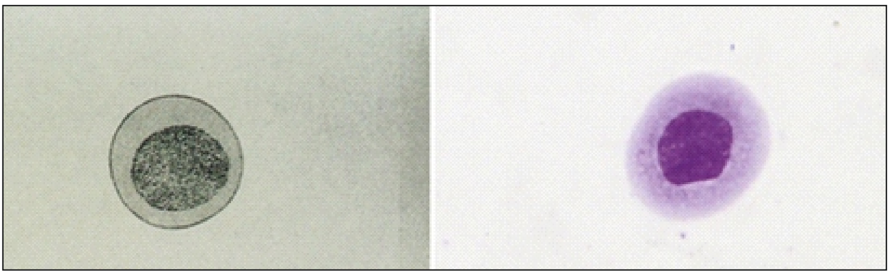
37. VAGINAL EXFOLIATIVE CYTOLOGY IN CANINES

CLASSIFICATIONS OF VAGINAL CELLS

The different cell types represent stages of cell death. As healthy round vaginal cells die, they then become larger and more irregular in shape. The nuclei within vaginal epithelial cells also undergo changes that reflect cell death. The nucleus becomes progressively smaller, then pyknotic, before disintegrating, leaving a nuclear "ghost" and then anuclear cell.

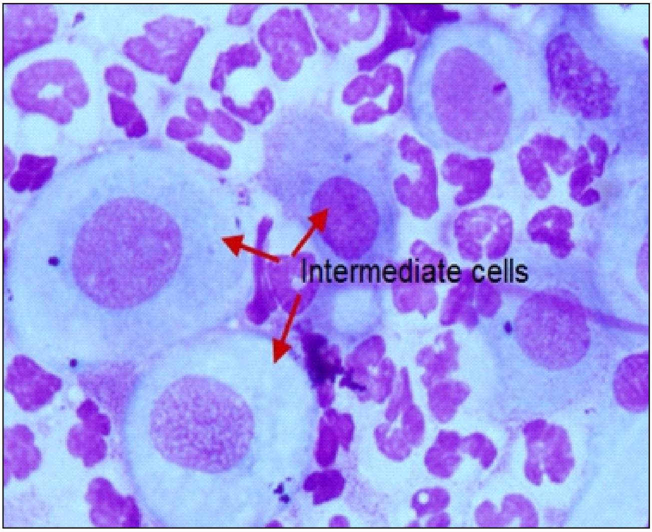
Parabasal Cells

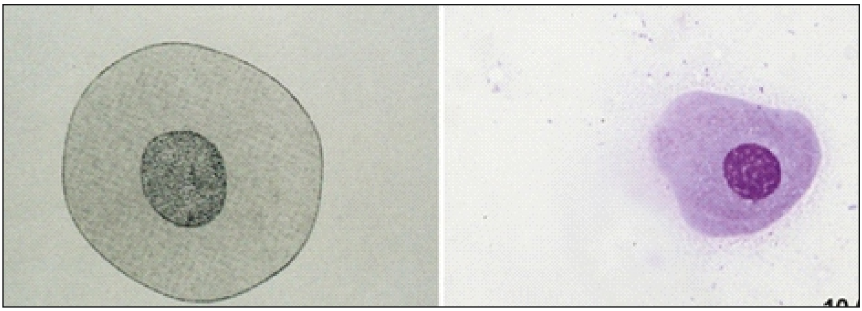




- Healthiest and smallest of vaginal cells.
- Round or slightly oval with large vesiculated nucleus, relatively small amount of cytoplasm.

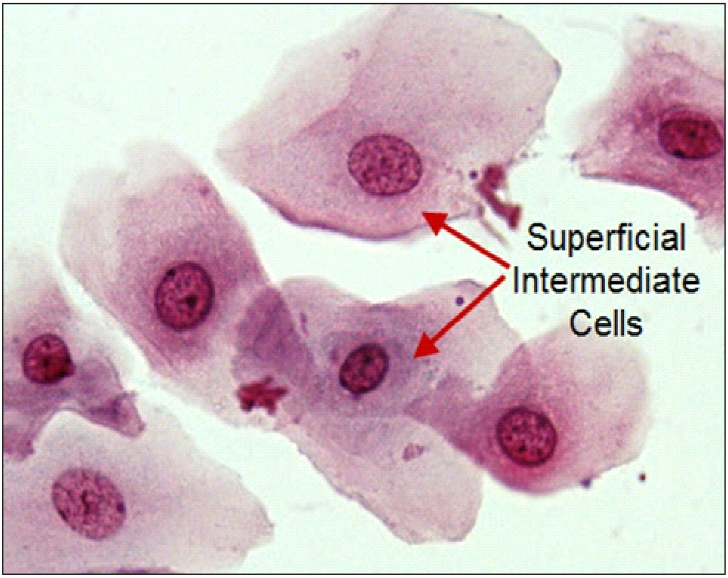
Intermediate cells

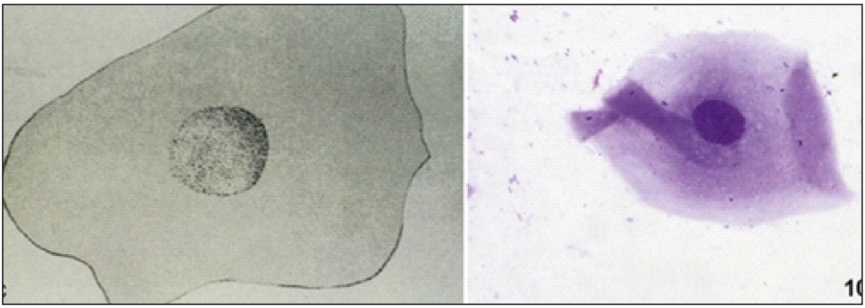




- Vary in size - slightly larger than parabasal cells to twice their size.
- Cells have smooth, oval to rounded irregular borders, vesiculated nucleus which is smaller than those found in parabasal cells.
- Change in cell morphology reflects first stage in cell death.
- Cells are larger with large amounts of cytoplasm and smaller nuclei.
- Described as small intermediate cells and large intermediate cells.

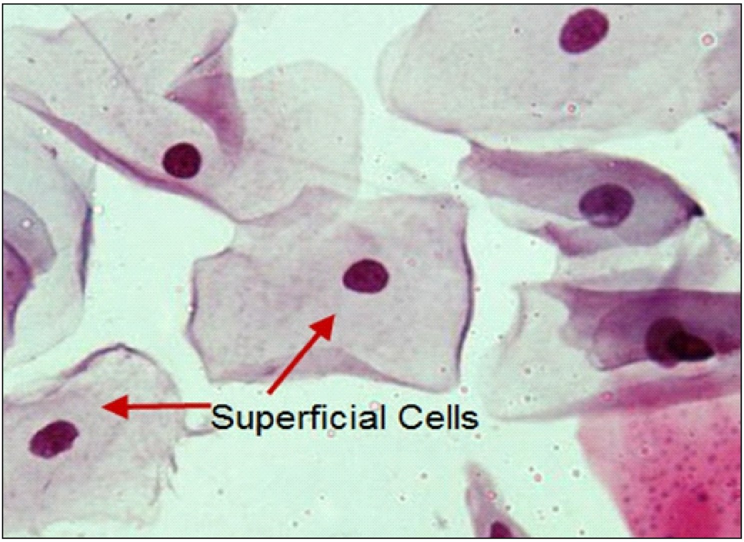
Superficial Intermediate cells

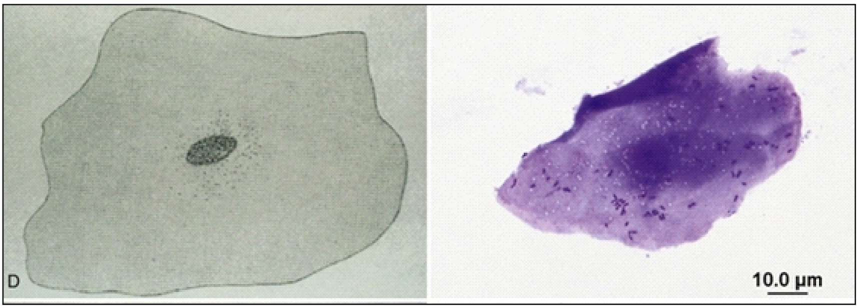




- Cells have angular, sharp, flat cytoplasmic border typical of superficial cells but the nucleus is still vesiculated.

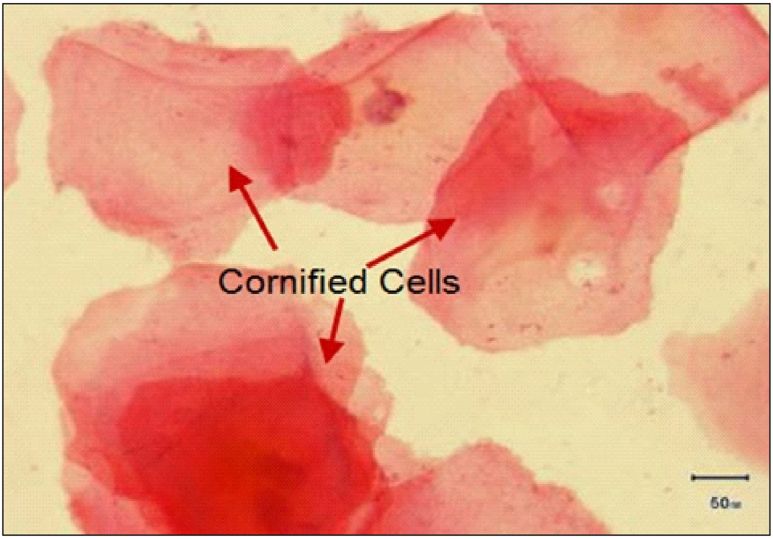
Superficial cells

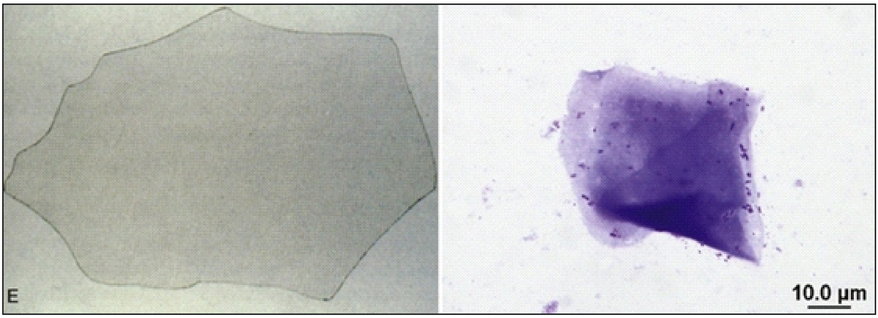




- Largest cells identified on vaginal cytology.
- Cells have sharp, flat, angular cytoplasmic borders and small pyknotic fading nuclei

Cornified cells





- Large, dead, irregular vaginal cells with no nucleus.
- These cells represent the end of the process that began with healthy round parabasal cells. These cells are also called as anuclear superficial cells, fully cornified or fully keratinized cells.

Interpretation

Stage	Red Blood Cells	Neutrophils	Bacteria	Epithelial Cells
Early Proestrus	Usually	Often	Many	Parabasal, intermediate, superficial intermediate, few superficial cells
Late Proestrus	Usually	Few or none	Many	Superficial Intermediate and Superficial
Estrus	Present, may be decreased in number or may be absent	None	Many	> 80-90% superficial or cornified cells
Diestrus	Some to none	Many	May be clearing	Parabasal, Intermediate, decreased superficial
Anoestrus	None	Few or none	None	Poor cellularity Parabasal and Intermediate cells

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